

The metric Blaschke theorem

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Abstract

We prove the metric Blaschke conjecture: every connected closed smooth Riemannian manifold of positive dimension whose global injectivity radius equals its diameter is, up to scale, a standard compact rank-one symmetric space.

1 Introduction

On a round sphere, every geodesic minimizes from its starting point to the antipode, at a distance equal to the diameter. The standard real, complex and quaternionic projective spaces and the Cayley plane have the same common minimizing-distance property. The metric Blaschke conjecture asks whether this property characterizes these compact rank-one symmetric spaces.

For a connected closed smooth Riemannian manifold (M, g) , write

$$\text{inj}(M, g) = \inf_{p \in M} \text{inj}_p(M, g), \quad \text{diam}(M, g) = \sup_{p, q \in M} d_g(p, q).$$

Here $\text{inj}_p(M, g)$ is the supremum of the radii on whose open tangent balls $\exp_p$ is a diffeomorphism onto its image. We call M a *Blaschke manifold* if it has positive dimension and $\text{inj}(M, g) = \text{diam}(M, g)$. The global equality implies that every unit-speed geodesic minimizes precisely up to the common diameter; we give the short deduction in Lemma 2.1.

Theorem 1.1 (The metric Blaschke theorem). *Let (M, g) be a connected closed smooth Riemannian manifold of positive dimension. If $\text{inj}(M, g) = \text{diam}(M, g)$, then (M, g) is isometric, up to multiplication of the metric by a positive constant, to a round sphere, a standard real, complex or quaternionic projective space, or the Cayley plane.*

The classification includes every quaternionic dimension. Its hypothesis is the global equality just defined. Classical Blaschke structure associates to M a compact rank-one symmetric *model* M_0 by its cut-locus and integral cohomology type. Throughout the volume comparisons, the standard metric g_0 on M_0 is scaled to have the same diameter as (M, g) . Prime geodesics have length twice the diameter, so fixing the diameter is equivalent to fixing their prime period.

1.1 The problem and its predecessors

Blaschke's original question concerned surfaces with a common reconvergence property for geodesics. Green settled the surface problem [4, 8]; the later geometric formulation and its development are described in Besse [3] and in the corrected account of McKay [14]. The classical theory connects the minimizing condition to closed geodesics, smooth cut loci and fibrations of tangent spheres by great spheres. The topological restrictions of Bott and Samelson explain the appearance of the compact rank-one symmetric models [3, Theorem 7.2].

The sphere and real-projective metric cases were established through the work of Berger, Kazdan, Weinstein and Yang; see Besse [3, Appendices D–E] and McKay [14, Sections 11–12]. Berger's sphere argument combines the geometric volume problem with Kazdan's scalar integral

inequality [1, 11]. Berger subsequently proved the sharp isoembolic inequality by a weighted comparison of Jacobi determinants and Jensen’s inequality [2, pp. 263–264]. These comparisons are central predecessors of the analytic part of the present proof.

Under additional hypotheses on families of totally geodesic round spheres of curvature 4, Rovenskii and Toponogov [20, Theorem 26, pp. 8–11] obtain a sharp projective volume comparison. Their Jacobi determinant splits into an exact curvature-4 factor and a complementary factor controlled by Kazdan’s inequality. Our fixed-chart Schur-complement argument uses the cut-time intersection pattern without requiring that geometric splitting.

For the projective models, it is useful to distinguish three kinds of information: topology, volume at a fixed geodesic period, and metric isometry. Yang proved that a Blaschke manifold of complex projective *homotopy type* has the model volume at the same prime geodesic period [28, Theorem, p. 380]. For the projective planes, smooth incidence geometry and the classification of Kramer and Stolz [13, Corollary C] identify the point manifold diffeomorphically. Wilking records the equal-volume theorem for a manifold homeomorphic to a compact rank-one symmetric space whose unit geodesics all have the same least period, attributing it to Reznikov [27, p. 282]. Reznikov’s earlier work on the quaternionic weak Blaschke conjecture [19] is another predecessor of this volume approach. The homeomorphism hypothesis in the result recorded by Wilking, and the other hypotheses just stated, will be verified at their uses. In higher quaternionic dimensions our upper bound is intrinsic: it is proved from the Blaschke geometry and integral topology in Sections 6–9.

1.2 Two volume bounds and their common equality case

The first part of the proof establishes a comparison valid for every model type.

Theorem 1.2 (Sharp volume comparison). *Every Blaschke manifold and its model at the same diameter satisfy*

$$\text{Vol}(M, g) \geq \text{Vol}(M_0, g_0).$$

Equality holds if and only if (M, g) is isometric to (M_0, g_0) .

Corollary 1.3 (Reduction to an upper volume bound). *If a Blaschke manifold has volume at most that of its model at the same diameter, then it is isometric to that model.*

After the circle, sphere and real-projective cases are separated, the analytic comparison in each remaining type starts along one closed geodesic. Normal Jacobi fields that vanish at a specified time form a Lagrangian plane in the symplectic space of Jacobi fields. A fixed graph chart separates the determinant of their endpoint pairing into a regular factor and a Schur-complement factor. The size of the Schur block is the dimension of the space of normal Jacobi fields vanishing at both the starting point and its cut point; the regular block has the complementary normal dimension. Both determinant factors can be bounded by scalar interval integrals, including when the actual endpoint order is reversed. This construction uses the cut-time intersection pattern; it does not require parallel eigenspaces of the curvature operator.

The resulting scalar estimates are combined by a logarithmic form of Kazdan’s weighted cancellation [11, Appendix E, especially E.4–E.8]. Its reflection symmetry cancels the integrated endpoint terms against interval averages. The scalar reference profiles are reciprocal-square sines, one on the full geodesic period and the other on a half-period. Folding the latter factor onto the two half-periods produces four weights, one for each choice of a half-period for each endpoint. Their equal, symmetric row and column sums allow both comparisons to contribute with the correct multiplicities. A nonnegative strict-convexity remainder records equality. Integrating over the unit tangent bundle gives the volume lower bound. Equality makes every radial volume density equal to its model value, and Szabó’s classification of compact simply connected harmonic manifolds then gives metric rigidity [24, 25].

The quaternionic upper bound has a different source. Fix dimension $4r$, with $r \geq 2$. Pairs of points at the cut distance form a smooth incidence manifold. Its actual normal four-plane bundle identifies the great-three-sphere families of tangent directions. Two different direction planes at one point intersect only at zero. The rank of their complement forces a formula for the tangent Pontryagin class; the signature theorem and a uniform estimate in every rank then determine its coefficient. Characteristic-class and signature theory enter here in their usual integral and oriented forms [16].

Next consider the space of oriented geodesics, forgetting the marked starting point. The period-normalized contact form expresses volume as a top power of the Euler class of its geodesic-circle bundle. Integral Gysin sequences first determine the lattice of classes and the determinant of a middle pairing. The first negative bundle of the free-loop energy transfers the incidence coefficient to this space. This step uses the fixed-endpoint Morse index theorem of Milnor [15] and the equivariant negative-bundle attachment developed in closed-geodesic Morse theory; we use the formulation of Ottosen and Bökstedt [18, Section 7], following Klingenberg [12]. The required primitive attaching coefficient is established over every prime field. The attaching map from the sphere bundle of the descended negative bundle to M also pulls back its cohomology generator; the vanishing of the $(r + 1)$ st power supplies the relations for the final algebraic calculation. Finally, a complete-intersection calculation turns the middle determinant into the sharp upper volume bound. Rational cohomology alone would not retain the arithmetic that normalizes this bound.

Section 2 fixes the geometric conventions. Sections 3 and 4 prove the geodesic comparison, and Section 5 completes its global volume and equality argument. Sections 6 and 7 establish the quaternionic characteristic class. Sections 8 and 9 convert it to the upper bound. Section 10 checks the remaining classical inputs and completes the metric classification.

1.3 Recent related rigidity results

There are also rigidity theorems with additional geometric hypotheses. Shankar, Spatzier and Wilking treat an upper sectional-curvature bound [21, Proposition 2.1], and Su, Sun and Wang prove the corresponding normalized lower-curvature result [23]. Connell, Islam, Nguyen and Spatzier obtain rigidity under curvature bounds and agreement with a model away from a prescribed convex set [6].

Song’s September 2026 preprint proves Blaschke metric rigidity assuming an adapted complex structure on the entire tangent bundle [22, Theorem 1.3]. This entire-extension hypothesis is additional to the Blaschke condition.

2 Blaschke geometry and model normalization

Lemma 2.1. *Let $D = \text{inj}(M, g) = \text{diam}(M, g)$. Every unit-speed geodesic segment of length at most D is minimizing. A segment of length less than D is the unique minimizing segment between its endpoints, and no segment of length greater than D is minimizing.*

Proof. Compactness and positive dimension give $0 < D < \infty$, and every pointwise injectivity radius is at least D . For $0 < t < D$, the exponential map at a point p is injective and nonsingular on a tangent ball containing the sphere of radius t . A minimizing geodesic to $\exp_p(tv)$ exists by completeness and has length at most t . Its initial vector lies in that ball and maps to the same endpoint as tv , so injectivity identifies the two vectors. This proves both minimality and uniqueness. Taking $t \uparrow D$ and using continuity of distance proves minimality at D . If a segment of length greater than D minimized, the distance between its endpoints would exceed the diameter. $\square$

In dimension one, a connected closed manifold is a circle. Arclength identifies its metric with a round circle of the same length. We henceforth suppose $n = \dim M \geq 2$.

We use the following classical structure conclusions [3, Chapters 5 and 7]. Precise statements in the corrected survey are McKay [14, Section 4, Corollary 1, Theorem 1, and Sections 7, 11–12].

Proposition 2.2 (Classical Blaschke structure). *Every unit-speed geodesic is simply closed with period $2D$. For each $p \in M$, the map $v \mapsto \exp_p(Dv)$ from $S_p M$ onto the cut locus is a smooth sphere fibration whose fibers are great spheres of one dimension k , independent of p . Its differential has nullity k . The model is determined as follows:*

- (i) *If $k = n - 1$, the model is a sphere; if $k = 0$, it is a real projective space. In these two cases the metric classification is known: the manifold is isometric to its equally scaled model.*
- (ii) *If $0 < k < n - 1$, the manifold is simply connected and its integral cohomology ring has one generator of degree $k + 1$. The possibilities are complex-projective type, quaternionic-projective type, and Cayley-plane type, with the dimensions listed below.*

The common period is the prime period, and each geodesic is simple. Indeed, if two distinct points of its parameter circle represented the same point of M , one of the two arcs between them would have length at most D and positive length. Lemma 2.1 would make that arc minimizing between equal endpoints, a contradiction. This also explains the freeness of the period action used later.

The simple-connectivity assertion in the intermediate case also has a short topological explanation. The homotopy sequence of $S^k \rightarrow S^{n-1} \rightarrow \text{Cut}(p)$ gives $\pi_1(\text{Cut}(p)) = 0$: the total sphere is simply connected and the fiber is connected. The exponential map identifies M with the space obtained by attaching a closed n -ball to its cut locus along this fibration. Indeed, it gives a continuous bijection from that compact quotient to the Hausdorff manifold M , hence a homeomorphism. Attaching this n -cell does not change the fundamental group, since the intermediate cases have $n \geq 4$.

Only the intermediate cases require the new comparison. From now on rescale the metric so that

$$D = \frac{\pi}{2}, \quad T = 2D = \pi, \quad d = n - 1, \quad m = d - k. \quad (2.1)$$

Both m and k are positive. The intermediate model data are

model	n	k	m
$\mathbb{CP}^r$ ($r \geq 2$)	$2r$	1	$2r - 2$
$\mathbb{HP}^r$ ($r \geq 2$)	$4r$	3	$4r - 4$
$\mathbb{OP}^2$	16	7	8

The projective lines are already included in the sphere case.

On a standard model at this scale, the normal radial curvature operator has parallel eigenspaces of dimensions m and k , with eigenvalues 1 and 4. Solving the scalar Jacobi equations with initial value zero and derivative one gives the polar density

$$w(t) = (\sin t)^m \left| \frac{\sin 2t}{2} \right|^k, \quad 0 \leq t \leq T. \quad (2.2)$$

It is positive away from 0, D , T and satisfies $w(T - t) = w(t)$. Let $\sigma_d = |S^d|$ denote ordinary unit-sphere area. The model volume is

$$V_0 = \sigma_d \int_0^D w(t) dt. \quad (2.3)$$

The eigenspace description is used only to calculate this model quantity. No curvature splitting or curvature bound is imposed on M .

Write $U = SM$, let ϕ_t be its geodesic flow, and use Liouville measure with unnormalized angular area:

$$d\mu(\xi) = d\text{vol}_g(p) d\sigma_p(\xi), \quad \xi \in S_p M. \quad (2.4)$$

This measure is invariant under the flow and has full support; if $V = \text{Vol}(M, g)$, then $\mu(U) = \sigma_d V$. Proposition 2.2 gives $\phi_T = \text{Id}_U$. Below, all determinants of Jacobi tensors are taken in absolute value, including beyond the cut time.

3 A Lagrangian comparison along a closed geodesic

We work in the intermediate case

$$D = \frac{\pi}{2}, \quad T = 2D = \pi, \quad d = n - 1, \quad 0 < k < d, \quad m = d - k.$$

Fix a unit-speed geodesic $\gamma : \mathbb{R} \rightarrow M$. The structural conclusions in Proposition 2.2 give the common period T , the absence of conjugate points at separations in $(0, D)$, and conjugate multiplicity k at separation D . This section uses these geometric conclusions and the Jacobi equation. Its comparison requires no assumption about the volume of M .

The purpose of the construction is to separate a Jacobi determinant into two factors with different scalar comparison profiles. Unlike the curvature-invariant splitting supplied by the totally geodesic curvature-four sphere families of Rovenskii and Toponogov [20, Theorem 26], this separation takes place in a fixed coordinate chart on the space of Jacobi fields. No splitting of the curvature operator is assumed.

3.1 The symplectic curve of vanishing Jacobi fields

We use the classical symplectic description of Jacobi fields; see, for example, Zelditch [29, Sections 2.1.2 and 10.5.2] and the variational-index framework of Duistermaat [7]. We derive the required identities to fix the signs and the normalization of the endpoint pairing.

Let $\mathcal{J}$ be the real vector space of normal Jacobi fields along γ . A prime on a field denotes covariant differentiation along γ . Prescribing the normal pair $(J(t), J'(t))$ at one time identifies $\mathcal{J}$ with a vector space of dimension $2d$. The form

$$\omega(I, J) = \langle I, J' \rangle - \langle I', J \rangle \quad (3.1)$$

is independent of the evaluation time: its derivative vanishes by the Jacobi equation and the self-adjointness of $X \mapsto R(X, \dot{\gamma})\dot{\gamma}$. Evaluation of the initial data identifies it with the standard nondegenerate symplectic form.

For each $t \in \mathbb{R}$, set

$$L_t = \{J \in \mathcal{J} : J(t) = 0\}.$$

The Jacobi initial-value problem shows that L_t depends smoothly on t , has dimension d , and is isotropic for ω . It is therefore Lagrangian. Its motion carries a natural positive inner product. For $I, J \in L_t$, choose a smooth local section $J_s \in L_s$ with $J_t = J$, and define

$$G_t(I, J) = \omega(I, \dot{J}_t), \quad \dot{J}_t = \left. \frac{d}{ds} \right|_{s=t} J_s \in \mathcal{J}.$$

The definition is independent of the extension: a section of L_s vanishing as an element of $\mathcal{J}$ at $s = t$ has derivative in L_t , which pairs to zero with I . Differentiating $J_s(s) = 0$ gives $\dot{J}_t(t) = -J'(t)$. Since $I(t) = 0$, our sign convention yields

$$G_t(I, J) = \langle I'(t), J'(t) \rangle. \quad (3.2)$$

Thus G_t is symmetric and positive definite, and $J \mapsto J'(t)$ is an isometry from (L_t, G_t) to the normal space $\dot{\gamma}(t)^\perp$.

We will need periodicity of the fields themselves. For any normal initial data (a, b) at time zero, the connection splitting of $T(SM)$ realizes (a, b) as the derivative of a smooth curve of unit initial velocities. The associated variation through unit-speed geodesics has Jacobi field with initial data (a, b) ; it is normal because its tangential component and its derivative vanish initially. All geodesics in this variation satisfy the same identity $\phi_T = \text{Id}_{SM}$. Differentiating that identity, or differentiating the equality of the varied positions and velocities after time T , gives

$$J(t+T) = J(t), \quad J'(t+T) = J'(t) \quad (J \in \mathcal{J}). \quad (3.3)$$

These are equalities in the same tangent space, since $\gamma(t+T) = \gamma(t)$. They do not assert that parallel transport around γ is the identity. In particular,

$$(L_{t+T}, G_{t+T}) = (L_t, G_t).$$

The intersection pattern is consequently

$$\dim(L_x \cap L_y) = \begin{cases} 0, & 0 < y - x < T, \quad y - x \neq D, \\ k, & y - x = D. \end{cases} \quad (3.4)$$

For $0 < y - x < D$, this is exponential nonsingularity. At separation D , evaluation at $y = x + D$ of the fields vanishing at x identifies the kernel with the kernel of the angular differential of the cut-locus fibration, whose dimension is k . If $D < y - x < T$, a field in the intersection also vanishes at $x + T$ by (3.3); the separation from y to $x + T$ is less than D , so the field is zero.

3.2 The determinant and a fixed graph chart

Proposition 3.1 (Jacobi determinant normalization). *Choose G_x - and G_y -orthonormal frames $Z_x = (Z_{x,1}, \dots, Z_{x,d})$ and $Z_y = (Z_{y,1}, \dots, Z_{y,d})$ of L_x and L_y . Then*

$$f(x, y) = \left| \det(\omega(Z_{x,a}, Z_{y,b}))_{a,b=1}^d \right| \quad (3.5)$$

is independent of these choices. It is continuous, symmetric in its arguments, and separately T -periodic. It is the absolute normal Jacobi determinant at y of fields vanishing at x with orthonormal initial derivatives there. In particular, for $0 < y - x < D$ it is the polar volume density with respect to dt and ordinary sphere area at $\gamma(x)$.

Use the same orthonormal basis of $\dot{\gamma}(0)^\perp$ for the coordinates

$$(q, p) = (J(0), J'(0)) \in \mathbb{R}^d \oplus \mathbb{R}^d,$$

choosing its first k vectors to be the initial derivatives of a basis of $L_0 \cap L_D$. On

$$\mathcal{I} = (0, D) \cup (D, T)$$

there is a smooth symmetric matrix $A(t)$ such that $L_t = \{(q, A(t)q) : q \in \mathbb{R}^d\}$, and $A'(t) > 0$. For $x, y \in \mathcal{I}$,

$$f(x, y) = \frac{|\det(A(y) - A(x))|}{\sqrt{\det A'(x) \det A'(y)}}. \quad (3.6)$$

Proof. Changing either orthonormal frame multiplies the pairing matrix by an orthogonal matrix. This leaves its absolute determinant unchanged. Local smooth orthonormal frames exist, so f is continuous even when the pairing is singular. Skew-symmetry of ω replaces the matrix by its negative transpose when the arguments are exchanged, proving symmetry. Periodicity follows from (3.3) and (3.2).

Evaluation of the pairing at y gives

$$\omega(Z_{x,a}, Z_{y,b}) = \langle Z_{x,a}(y), Z'_{y,b}(y) \rangle.$$

The vectors $Z'_{y,b}(y)$ form an orthonormal normal basis, and the fields $Z_{x,a}$ have zero values and orthonormal derivatives at x . This proves the claimed determinant interpretation. More explicitly, an angular variation of a unit direction with derivative e gives the Jacobi field J with $J(x) = 0$ and $J'(x) = e$. Its value at $x + r$ is the angular derivative of $\exp_{\gamma(x)}(r \cdot)$. Together with the unit radial derivative and Gauss's lemma, this gives the polar density $f(x, x + r) dr d\sigma$. There is no additional power of r in this polar formula; the Cartesian normal-coordinate density would be $f(x, x + r)/r^d$.

In the stated coordinates,

$$\omega((q, p), (\tilde{q}, \tilde{p})) = q^\top \tilde{p} - p^\top \tilde{q}, \quad L_0 = \{q = 0\}.$$

Equation (3.4) makes L_t transverse to L_0 for $t \in \mathcal{I}$, proving the graph representation. Isotropy is equivalent to $A = A^\top$. For the graph frame $E_t q = (q, A(t)q)$,

$$G_t(E_t q, E_t r) = q^\top A'(t)r, \quad \omega(E_x q, E_y r) = q^\top (A(y) - A(x))r.$$

Positivity of G_t gives $A'(t) > 0$. The frames $E_t A'(t)^{-1/2}$ are orthonormal for G_t , so taking determinants in the last identity proves (3.6). $\square$

The coordinate basis is now fixed. Split both q and p into their first k and last m coordinates and write

$$A(t) = \begin{pmatrix} V_b(t) & C(t) \\ C(t)^\top & H(t) \end{pmatrix}, \quad S(t) = V'_b(t) - C'(t)H'(t)^{-1}C'(t)^\top. \quad (3.7)$$

The subscript in V_b distinguishes this matrix from the volume of the manifold. Here H is an $m \times m$ matrix and S is a $k \times k$ matrix. Positive definiteness of A' and block elimination give

$$H'(t) > 0, \quad S(t) > 0, \quad \det A'(t) = \det H'(t) \det S(t) \quad (t \in \mathcal{I}). \quad (3.8)$$

For comparison, on the standard model the adapted chart has

$$A_{\text{mod}}(t) = \text{diag}(-2 \cot(2t) \text{Id}_k, -\cot(t) \text{Id}_m).$$

Thus $H'_{\text{mod}}(t) = \csc^2(t) \text{Id}_m$ and $S_{\text{mod}}(t) = 4 \csc^2(2t) \text{Id}_k$. These formulas motivate the scalar reference profiles below.

3.3 The chart poles and determinant-root profiles

The curve L_t remains smooth at $0, D, T$; it is the projection from L_t to the fixed q -coordinate space that ceases to be invertible. Its kernel has dimension d at $0, T$ and dimension k at D . The following expansion shows how this loss of transversality enters A : at D the pole lies entirely in the first k coordinates, while the complementary block H remains regular. Smooth remainders are needed because the comparison uses derivatives of the graph matrices.

Lemma 3.2 (Pole expansions). *There is a positive definite $k \times k$ matrix B such that the graph matrix has the expansions*

$$A(t) = -\frac{1}{t-D} \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} + R_D(t), \quad \text{near } D, \quad (3.9)$$

$$A(t) = -\frac{1}{t-t_0} \text{Id}_d + R_{t_0}(t), \quad \text{near } t_0 = 0, T, \quad (3.10)$$

where each remainder extends smoothly across its indicated point. In particular, C and H extend smoothly across D , and $H'(D) > 0$.

Proof. Put $\tau = t - D$. Choose a local smooth frame of L_t , written as the columns of

$$F(t) = \begin{pmatrix} X(t) \\ P(t) \end{pmatrix},$$

whose first k columns at D are the vertical vectors $(0, e_a)$ for $1 \leq a \leq k$. These are exactly the chosen coordinate basis of $L_0 \cap L_D$. Isotropy with these vectors forces every vector in the image of $X(D)$ to have its first k coordinates zero. The kernel of the value projection on L_D is $L_0 \cap L_D$, so that image has dimension m and is the entire last coordinate space. Choose the remaining frame columns so that their value projections give a basis of this image. In blocks, their last m value coordinates form an invertible matrix M .

The first k columns of $X(D)$ vanish. Dividing these columns by τ gives a smooth matrix $\tilde{X}(t)$ satisfying

$$X(t) = \tilde{X}(t) \begin{pmatrix} \tau \text{Id}_k & 0 \\ 0 & \text{Id}_m \end{pmatrix}.$$

Let G be the matrix of G_D on the first k frame vectors. For $1 \leq a, b \leq k$, the definition of the velocity form in the fixed coordinates gives

$$G_{ab} = \omega((0, e_a), \dot{F}_b(D)) = -e_a^\top X'_b(D).$$

Consequently

$$\tilde{X}(D) = \begin{pmatrix} -G & 0 \\ N & M \end{pmatrix}, \quad G > 0, \quad M \text{ invertible}, \quad (3.11)$$

for some matrix N . In particular $\tilde{X}(D)$ is invertible. For $\tau \neq 0$ sufficiently small,

$$A(t) = P(t) \begin{pmatrix} \tau^{-1} \text{Id}_k & 0 \\ 0 & \text{Id}_m \end{pmatrix} \tilde{X}(t)^{-1}.$$

To identify its residue, let $P_1(t)$ denote the first k columns of $P(t)$ and let $U_1(t)$ denote the first k rows of $\tilde{X}(t)^{-1}$. Then the only possibly singular summand is $\tau^{-1} P_1(t) U_1(t)$. At D ,

$$P_1(D) = \begin{pmatrix} \text{Id}_k \\ 0 \end{pmatrix}, \quad U_1(D) = \begin{pmatrix} -G^{-1} & 0 \end{pmatrix}.$$

Their product is $-\text{diag}(G^{-1}, 0)$. The difference $P_1(t) U_1(t) - P_1(D) U_1(D)$ divided by τ is smooth, as is the summand from the other frame columns. This proves (3.9) with $B = G^{-1} > 0$, including the absence of any off-diagonal residue.

For fixed $z \in \mathbb{R}^m$, the graph vectors

$$E_z(t) = \left(\begin{pmatrix} 0 \\ z \end{pmatrix}, \begin{pmatrix} C(t)z \\ H(t)z \end{pmatrix} \right)$$

extend smoothly through D . Their limits lie in L_D by continuity of L_t , and are nonzero if $z \neq 0$. The velocity identity in the fixed symplectic coordinates, now also at D , gives

$$z^\top H'(D) z = \omega(E_z(D), \dot{E}_z(D)) = G_D(E_z(D), E_z(D)) > 0.$$

This proves strict positivity of the extended derivative.

At $t_0 = 0$, take a frame whose value at 0 is the whole vertical coordinate basis. All X -columns then vanish at 0. The velocity metric on this basis is Id_d by (3.2); hence $X'(0) = -\text{Id}_d$ and $P(0) = \text{Id}_d$. Dividing all value columns by t and repeating the preceding calculation gives residue $-\text{Id}_d$. At $t_0 = T$, periodicity identifies both L_T and its velocity metric with their values at zero in these same fixed coordinates. The same argument gives residue $-\text{Id}_d$ there, proving (3.10). $\square$

Proposition 3.3 (Scalar profiles). *Define*

$$\begin{aligned} h(t) &= \det(H'(t))^{1/m} & (0 < t < T), \\ s(t) &= \det(S(t))^{1/k} & (t \in \mathcal{I}), \\ s_i(t) &= s(t + iD) & (0 < t < D, \ i = 0, 1). \end{aligned} \quad (3.12)$$

The function h is smooth and positive across D . There are positive constants c_H, C_H, c_i, C_i , depending on the fixed geodesic and its chart, such that

$$c_H \leq h(t) \sin^2 t \leq C_H \quad (0 < t < T), \quad (3.13)$$

$$c_i \leq \frac{s_i(t) \sin^2(2t)}{4} \leq C_i \quad (0 < t < D, \ i = 0, 1). \quad (3.14)$$

Equivalently, the logarithms of the ratios of h to $1/\sin^2 t$ and of s_i to $4/\sin^2(2t)$ are bounded.

Proof. Lemma 3.2 and (3.8) show that H' is positive throughout $(0, T)$, including D . Near $t_0 = 0, T$, put $\tau = t - t_0$. Differentiating the smooth-remainder expansion gives

$$H'(t) = \tau^{-2} \text{Id}_m + O(1), \quad V'_b(t) = \tau^{-2} \text{Id}_k + O(1), \quad C'(t) = O(1).$$

Here and below $O(1)$ denotes a matrix bounded in operator norm as the indicated time is approached. Since $H'(t)^{-1} = O(\tau^2)$, the Schur complement satisfies

$$S(t) = \tau^{-2} \text{Id}_k + O(1) \quad (t \rightarrow 0, T). \quad (3.15)$$

At D , both C' and H' are smooth and $H'(D)$ is invertible. The middle pole therefore yields

$$S(t) = (t - D)^{-2} B + O(1) \quad (t \rightarrow D). \quad (3.16)$$

Taking determinant roots gives the positive limits

	left endpoint	right endpoint
$h(t) \sin^2 t \quad (0 < t < T)$	1	1
$s_0(t) \sin^2(2t)/4 \quad (0 < t < D)$	1	$\det(B)^{1/k}$
$s_1(t) \sin^2(2t)/4 \quad (0 < t < D)$	$\det(B)^{1/k}$	1.

Each ratio thus extends continuously and positively to its compact closed interval. Its minimum and maximum supply the stated bounds. $\square$

3.4 Schur complements for arbitrary endpoint order

The Schur complement and determinant-root concavity are standard matrix tools [5, Appendix A.5.5 and Exercises 3.18(b), 3.26(b)]. Here the essential point is to apply the factorization with either actual endpoint order. We retain the matrix calculation so that its signs and integration paths are explicit.

For distinct $x, y \in \mathcal{I}$, use Δ to denote the difference of the values at y and x . Since H is smooth and $H' > 0$ throughout $(0, T)$,

$$\Delta H = H(y) - H(x) = \int_x^y H'(t) dt \quad (3.17)$$

is positive definite for $y > x$ and negative definite for $y < x$. In particular it is invertible in either order. Define

$$Q(x, y) = \Delta V_b - (\Delta C)(\Delta H)^{-1}(\Delta C)^T. \quad (3.18)$$

Thus Q is the Schur complement of the regular block in the endpoint difference ΔA , whereas S in (3.7) is the Schur complement in the velocity A' . The next proposition relates them by differentiating in either endpoint; the diagonal and cut-pair zeros of Q then anchor the two interval bounds.

Proposition 3.4 (Two Schur-complement bounds). *The matrix $Q(x, y)$ is symmetric and satisfies*

$$\det(\Delta A) = \det(\Delta H) \det Q(x, y), \quad Q(y, x) = -Q(x, y). \quad (3.19)$$

It extends continuously by zero to $y = x \in \mathcal{I}$, and

$$Q(u, u + D) = Q(u + D, u) = 0 \quad (0 < u < D). \quad (3.20)$$

Whenever its distinct endpoints remain in $\mathcal{I}$, it obeys the positive-semidefinite matrix inequalities

$$\partial_y Q(x, y) \geq S(y), \quad -\partial_x Q(x, y) \geq S(x). \quad (3.21)$$

Consequently, for every $0 < u < v < D$ and $i, j \in \{0, 1\}$,

$$Q(u + iD, v + jD) \geq \int_u^v S(t + jD) dt, \quad (3.22)$$

$$Q(u + iD, v + jD) \geq \int_u^v S(t + iD) dt. \quad (3.23)$$

In particular, this Q is positive definite for all four choices of i, j , including $i = 1, j = 0$.

Proof. With $R = (\Delta C)(\Delta H)^{-1}$, block congruence gives

$$\begin{pmatrix} \text{Id}_k & -R \\ 0 & \text{Id}_m \end{pmatrix} (\Delta A) \begin{pmatrix} \text{Id}_k & 0 \\ -R^\top & \text{Id}_m \end{pmatrix} = \begin{pmatrix} Q & 0 \\ 0 & \Delta H \end{pmatrix}. \quad (3.24)$$

This proves the determinant identity and identifies the nullity of ΔA with that of Q . Exchanging the endpoints changes the sign of each difference, and hence changes Q to $-Q$.

For a matched pair $(u, u + D)$, the kernel of ΔA parametrizes $L_u \cap L_{u+D}$ by the common q coordinate. It has dimension k by (3.4). The matrix Q has size $k \times k$ and therefore has full nullity: it is the zero matrix. This proves (3.20), also in the reversed order. Near a diagonal point within the chart, Taylor expansion gives

$$\Delta H = (y - x)H'(x) + O((y - x)^2), \quad \Delta C = (y - x)C'(x) + O((y - x)^2),$$

and the corresponding expansion for ΔV_b . Invertibility of $H'(x)$ then gives

$$Q(x, y) = (y - x)S(x) + O((y - x)^2). \quad (3.25)$$

This establishes the zero extension needed at the endpoints of the integrations below.

Direct differentiation of (3.18), using $\partial_y(\Delta H)^{-1} = -(\Delta H)^{-1}H'(y)(\Delta H)^{-1}$, gives

$$\begin{aligned} \partial_y Q &= V'_b(y) - C'(y)R^\top - RC'(y)^\top + RH'(y)R^\top \\ &= \begin{pmatrix} \text{Id}_k & -R \end{pmatrix} A'(y) \begin{pmatrix} \text{Id}_k \\ -R^\top \end{pmatrix}. \end{aligned}$$

The identical calculation with $-\partial_x$ gives the same expression with $A'(x)$. For either endpoint $a = x$ or $a = y$, completing the square gives the exact identity

$$\begin{aligned} &\begin{pmatrix} \text{Id}_k & -R \end{pmatrix} A'(a) \begin{pmatrix} \text{Id}_k \\ -R^\top \end{pmatrix} \\ &= S(a) + (R - C'(a)H'(a)^{-1})H'(a)(R - C'(a)H'(a)^{-1})^\top. \end{aligned} \quad (3.26)$$

The second summand is positive semidefinite. This proves (3.21). The calculation used invertibility of ΔH , but did not require its positive definiteness.

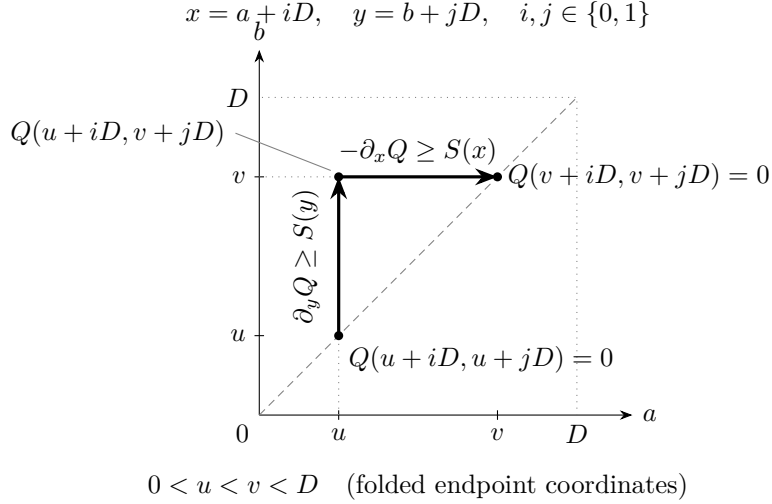


Figure 1: The two paths in folded endpoint coordinates $x = a + iD, y = b + jD$, for $0 < u < v < D$. The first starts at $Q = 0$ and increases the second endpoint; the second increases the first endpoint toward another zero. The zeros come from the diagonal when $i = j$ and from the cut-pair identity otherwise. Each moving endpoint stays in one open half-period. When $i = 1, j = 0$, the actual order is $u + D > v$, and both positive-semidefinite lower bounds still apply.

To obtain (3.22), fix the first endpoint $u + iD$ and move the second from $u + jD$ to $v + jD$. The starting value of Q is zero by the diagonal extension if $i = j$, and by (3.20) otherwise. Integrating $\partial_y Q \geq S(y)$ on this interval gives the result. For (3.23), fix the second endpoint $v + jD$ and move the first from $u + iD$ to $v + iD$. The terminal value is zero, so integration of $-\partial_x Q \geq S(x)$ gives the second result. When $i = j$, one can first integrate on a shortened interval and pass to the diagonal endpoint using (3.25).

Figure 1 displays these integrations with the half-period indices held fixed.

Both moving endpoints stay in one open half-period, so neither path passes through a chart pole. In particular, when $i = 1, j = 0$ the actual endpoint order is $u + D > v$. The first path starts at $(u + D, u)$ and the second ends at $(v + D, v)$; the same zero values and the same derivative inequalities apply. Thus neither matrix inequality reverses in this case. Finally, the integrals on their right sides are positive definite, so $Q(u + iD, v + jD) > 0$. $\square$

3.5 Scalar lower bounds for the two factors

For distinct $x, y \in \mathcal{I}$, define

$$f_H(x, y) = \frac{|\det(H(y) - H(x))|}{\sqrt{\det H'(x) \det H'(y)}}, \quad f_K(x, y) = \frac{|\det Q(x, y)|}{\sqrt{\det S(x) \det S(y)}}. \quad (3.27)$$

These auxiliary factors are symmetric under exchanging the actual endpoints. Combining (3.6), (3.8), and (3.19) gives

$$f(x, y) = f_H(x, y) f_K(x, y). \quad (3.28)$$

The factors are positive away from the cut pairs $|y - x| = D$; at those pairs $f_K = 0$.

Proposition 3.5 (Scalar block bounds). *With the profiles of Proposition 3.3, every pair of distinct chart endpoints satisfies*

$$f_H(x, y)^{1/m} \geq \frac{|\int_x^y h(t) dt|}{\sqrt{h(x)h(y)}}. \quad (3.29)$$

For $0 < u < v < D$ and $i, j \in \{0, 1\}$ one has

$$f_K(u + iD, v + jD)^{1/k} \geq \frac{(\int_u^v s_i(t) dt)^{1/2} (\int_u^v s_j(t) dt)^{1/2}}{\sqrt{s_i(u)s_j(v)}}. \quad (3.30)$$

Together with endpoint exchange, these bounds cover the full square $(0, T)^2$ except the chart, diagonal, and cut lines, a set of planar measure zero.

Proof. We first record the determinant inequality used in both blocks. For a positive definite $r \times r$ matrix M , write $\delta_r(M) = \det(M)^{1/r}$. The elementary variational formula

$$\delta_r(M) = \inf_{\substack{P \geq 0 \\ \det P = 1}} \frac{1}{r} \operatorname{tr}(PM) \quad (3.31)$$

follows by applying the arithmetic–geometric mean inequality to the eigenvalues of $M^{1/2}PM^{1/2}$. Equality is attained at $P = \delta_r(M)M^{-1}$. The formula proves monotonicity under the positive-semidefinite order. It also proves, for any continuous positive definite matrix function on a compact interval,

$$\delta_r \left(\int_a^b M(t) dt \right) \geq \int_a^b \delta_r(M(t)) dt. \quad (3.32)$$

Indeed, for every admissible P the integral of $\operatorname{tr}(PM(t))/r$ is at least the right side, and one then takes the infimum over P . This is the integral form of Minkowski’s determinant inequality; it retains the degree-one homogeneity of the determinant root.

Apply (3.32) to H' on the interval between x and y . Equation (3.17) gives

$$|\det(\Delta H)|^{1/m} \geq \left| \int_x^y \det(H'(t))^{1/m} dt \right|.$$

For $y < x$ this uses the positive matrix $-\Delta H$. Division by

$$(\det H'(x) \det H'(y))^{1/(2m)} = \sqrt{h(x)h(y)}$$

proves (3.29).

Now put $Q = Q(u + iD, v + jD) > 0$. Each of the two matrix bounds in Proposition 3.4, followed by determinant monotonicity and (3.32), gives

$$\delta_k(Q) \geq \int_u^v s_i(t) dt, \quad \delta_k(Q) \geq \int_u^v s_j(t) dt.$$

Thus $\delta_k(Q)$ dominates their maximum and therefore their geometric mean. The denominator in the k th root of f_K is exactly

$$(\det S(u + iD) \det S(v + jD))^{1/(2k)} = \sqrt{s_i(u)s_j(v)},$$

which proves (3.30) with the displayed powers.

Finally, every endpoint in $\mathcal{I}$ has a unique expression $u + iD$ with $0 < u < D$ and $i \in \{0, 1\}$. For a pair whose folded parameters differ, order those parameters as $u < v$, exchanging the actual endpoints if necessary. The preceding bounds apply for every choice of the two half-period indices. Equal folded parameters mean either a diagonal pair or a cut pair. The remaining excluded pairs have an endpoint at D . These are finitely many lines, and symmetry of both factors justifies the endpoint exchange everywhere else. $\square$

The two determinant roots have now been compared with scalar interval integrals. Proposition 3.3 supplies exactly their endpoint control. The next section combines these bounds by a weighted logarithmic identity.

4 Weighted comparison along a closed geodesic

We retain $D = \pi/2$, $T = \pi$, $d = n-1$, and $m = d-k$, with $m, k \geq 1$. The factorization $f = f_H f_K$ turns the logarithm of the ratio to the model density into a sum of two block contributions. We will show that each contribution has nonnegative integral with weight $w(|y-x|)$. The scalar identity below cancels the integrated endpoint terms in the block bounds. A strict-convexity remainder will then give the determinant comparison and identify its equality case, without separately classifying equality in the matrix estimates.

The scalar identity is a logarithmic form of the weighted cancellation in Kazdan's proof of the Berger–Kazdan inequality [11, Appendix E, especially E.4–E.8]. At interval length π , writing Kazdan's profile as $\phi(t) = \sin(t)e^{u(t)}$ gives $p = \phi^{-2}$, $p_0(t) = \sin^{-2} t$, $g = \log(p/p_0) = -2u$, and weight $\rho(t) = W(t)/\sin t$. We give the version needed here, with bounded logarithmic ratios and a measurable weight, including the endpoint integrability. Its application to the two factors depends on the folded symmetry proved below.

Lemma 4.1 (Weighted scalar identity). *Let $b > 0$, $\lambda = \pi/b$, and*

$$p_0(t) = \frac{\lambda^2}{\sin^2(\lambda t)}, \quad 0 < t < b.$$

Let $p : (0, b) \rightarrow (0, \infty)$ be continuous and suppose that $g = \log(p/p_0)$ is bounded. Let W be nonnegative and measurable on $(0, b)$, with $W(b-z) = W(z)$ almost everywhere and $W(z)/\sin(\lambda z)$ essentially bounded. Write

$$\Delta_b = \{(x, y) : 0 < x < y < b\}, \quad d\nu_{xy}(t) = \frac{p_0(t) dt}{\int_x^y p_0}, \quad x < t < y.$$

Then all the following integrals are finite, and

$$\begin{aligned} \int_{\Delta_b} W(y-x) \left[\log \frac{\int_x^y p}{\int_x^y p_0} - \frac{g(x) + g(y)}{2} \right] dx dy \\ = \int_{\Delta_b} W(y-x) \left[\log \int_x^y e^{g(t)} d\nu_{xy}(t) - \int_x^y g(t) d\nu_{xy}(t) \right] dx dy \geq 0. \end{aligned} \quad (4.1)$$

Proof. Direct integration gives

$$P_0(x, y) := \int_x^y p_0 = \frac{\lambda \sin(\lambda(y-x))}{\sin(\lambda x) \sin(\lambda y)}, \quad \frac{P_0(x, y)}{\sqrt{p_0(x)p_0(y)}} = \frac{\sin(\lambda(y-x))}{\lambda}. \quad (4.2)$$

The measure ν_{xy} is a probability measure. Both its average of g and $\log \int e^g d\nu_{xy} = \log(\int p / \int p_0)$ have absolute value at most $\|g\|_\infty$. Since W is essentially bounded, the asserted finiteness follows already from these bounds. Jensen's inequality makes the integrand on the second line of (4.1) nonnegative. It remains to establish the equality of the integrated linear terms.

Set $K(z) = W(z)/\sin(\lambda z)$ and $I_W = \int_0^b W(z) dz$, and define

$$F(t) = \int_{0 < x < t < y < b} K(y-x) \sin(\lambda x) \sin(\lambda y) dx dy.$$

Here the two-variable kernel $q(x, y) = K(y-x) \sin(\lambda x) \sin(\lambda y)$ is essentially bounded. Fubini gives

$$F(t) = \int_0^t \left(\int_x^b q(x, y) dy \right) dx - \int_0^t \left(\int_0^y q(x, y) dx \right) dy.$$

Thus F is absolutely continuous on $[0, b]$, $F(0) = 0$, and, for almost every t ,

$$\begin{aligned} F'(t) &= \sin(\lambda t) \left[\int_0^{b-t} K(z) \sin(\lambda(t+z)) dz - \int_0^t K(z) \sin(\lambda(t-z)) dz \right] \\ &= \sin(\lambda t) \int_0^b K(z) \sin(\lambda(t+z)) dz \\ &= I_W \sin(\lambda t) \cos(\lambda t). \end{aligned}$$

For the second equality substitute $b - z$ in the subtracted integral: $K(b - z) = K(z)$ and $\sin(\lambda(t - b + z)) = -\sin(\lambda(t + z))$. For the last equality, symmetry gives $\int_0^b K(z) \cos(\lambda z) dz = 0$, while $\int_0^b K(z) \sin(\lambda z) dz = I_W$. Integration therefore yields

$$F(t) = \frac{I_W}{2\lambda} \sin^2(\lambda t), \quad \frac{p_0(t)}{\lambda} F(t) = \frac{I_W}{2}. \quad (4.3)$$

This argument has not differentiated the measurable weight.

Using (4.2) and then (4.3), the integrated interval averages are

$$\begin{aligned} \int_{\Delta_b} W(y-x) \int_x^y g(t) d\nu_{xy}(t) dx dy &= \int_0^b g(t) \frac{p_0(t)}{\lambda} F(t) dt \\ &= \frac{I_W}{2} \int_0^b g(t) dt. \end{aligned}$$

The exchange is absolutely integrable: replacing g by $|g|$ in the left side bounds it by $\|g\|_\infty \int_{\Delta_b} W(y-x) dx dy$. In particular, no integral of p_0 over the entire interval is taken. The endpoint terms have exactly the same value, since

$$\begin{aligned} \frac{1}{2} \int_{\Delta_b} W(y-x)(g(x) + g(y)) dx dy \\ = \frac{1}{2} \int_0^b g(t) \left[\int_0^{b-t} W(z) dz + \int_0^t W(z) dz \right] dt = \frac{I_W}{2} \int_0^b g(t) dt. \end{aligned}$$

The final equality again uses $W(b-z) = W(z)$. Subtracting these equal finite quantities proves the identity and hence the lemma. $\square$

Theorem 4.2 (Comparison on each geodesic). *In the normalized intermediate case of Proposition 2.2, let $f(x, y)$ be the Jacobi determinant of any fixed unit-speed geodesic, as in Proposition 3.1. Put*

$$\mathcal{R} = (0, T)^2, \quad w(z) = (\sin z)^m \left| \frac{\sin(2z)}{2} \right|^k \quad (0 \leq z \leq T).$$

Then

$$\int_{\mathcal{R}} f(x, y) dx dy \geq \int_{\mathcal{R}} w(|y-x|) dx dy. \quad (4.4)$$

Equality holds if and only if $f(x, y) = w(|y-x|)$ for every $(x, y) \in [0, T]^2$.

Proof. All constants in this proof may depend on the chosen geodesic. Write $w_{xy} = w(|y-x|)$. We first work off the null set

$$E = \{(x, y) \in \mathcal{R} : x = D \text{ or } y = D \text{ or } x = y \text{ or } |y-x| = D\}.$$

There f, w_{xy} , and the two determinant factors f_H, f_K of Proposition 3.5 are positive, and $f = f_H f_K$. Define

$$L_H(x, y) = \log \frac{f_H(x, y)}{\sin^m(|y-x|)}, \quad L_K(x, y) = \log \frac{f_K(x, y)}{|\sin(2|y-x|)/2|^k}.$$

Set their values on E equal to zero for integration. Both are symmetric in the actual endpoint slots x, y .

Finiteness of the logarithmic integrals. The profile bounds in Proposition 3.3 give constants $0 < \alpha \leq \beta < \infty$ such that

$$\begin{aligned}\alpha &\leq \frac{h(t)}{\csc^2 t} \leq \beta & (0 < t < T), \\ \alpha &\leq \frac{s_i(t)}{4 \csc^2(2t)} \leq \beta & (0 < t < D, \ i = 0, 1).\end{aligned}$$

Set $\eta = \alpha/\beta \leq 1$. For $x < y$, the first bound in Proposition 3.5 and the chord identity (4.2) imply

$$f_H(x, y)^{1/m} \geq \frac{\int_x^y h}{\sqrt{h(x)h(y)}} \geq \eta \sin(y - x).$$

Symmetry gives this lower bound for either endpoint order. Similarly, put $B_0(z) = \sin(2z)/2$ for $0 < z < D$. For $0 < u < v < D$ and $i, j \in \{0, 1\}$, the second block bound gives

$$f_K(u + iD, v + jD)^{1/k} \geq \frac{(\int_u^v s_i)^{1/2} (\int_u^v s_j)^{1/2}}{\sqrt{s_i(u)s_j(v)}} \geq \eta B_0(v - u).$$

In these coordinates $B_0(v - u) = |\sin(2|y - x|)/2|$. Every pair outside E has this form, possibly after exchanging endpoints. Consequently

$$L_H \geq m \log \eta, \quad L_K \geq k \log \eta, \quad L_H + L_K = L := \log(f/w_{xy}). \quad (4.5)$$

With $a^+ = \max(a, 0)$, the elementary estimate

$$w_{xy}L^+ \leq f$$

and continuity of f on the compact square prove integrability of $w_{xy}L^+$. The individual negative parts are bounded by (4.5); also

$$L_H^+ \leq L^+ - k \log \eta, \quad L_K^+ \leq L^+ - m \log \eta.$$

Thus $w_{xy}L_H$, $w_{xy}L_K$, and $w_{xy}L$ are all absolutely integrable. The following additions of logarithmic integrals therefore involve only finite quantities.

The H factor on the full square. Apply Lemma 4.1 with $b = T$, $\lambda = 1$, $p = h$, and $W = w$. Its assumptions hold by Proposition 3.3 and

$$w(T - z) = w(z), \quad \frac{w(z)}{\sin z} = \sin^{m-1} z \left| \frac{\sin(2z)}{2} \right|^k,$$

which is bounded because $m \geq 1$. Writing $g_H = \log(h/\csc^2 t)$, the first block bound and (4.2) give, for $0 < x < y < T$ outside E ,

$$\frac{L_H(x, y)}{m} \geq \log \frac{\int_x^y h(t) dt}{\int_x^y \csc^2 t dt} - \frac{g_H(x) + g_H(y)}{2}.$$

After multiplication by $w(y - x)$, Lemma 4.1 makes the integral of the right side nonnegative. Multiplication by m and endpoint symmetry then yield

$$\int_{\mathcal{R}} w_{xy} L_H(x, y) dx dy \geq 0. \quad (4.6)$$

The K factor and the four folded weights. Put $p_0(t) = 4 \csc^2(2t)$ on $(0, D)$ and define

$$g_i(t) = \log \frac{s_i(t)}{p_0(t)}, \quad A_i(u, v) = \log \frac{\int_u^v s_i(t) dt}{\int_u^v p_0(t) dt} \quad (i = 0, 1).$$

For $(u, v) \in \Delta_D$, $z = v - u$, $x = u + iD$, and $y = v + jD$, the second block bound gives the precise logarithmic inequality

$$\frac{L_K(u + iD, v + jD)}{k} \geq \frac{1}{2} [A_i(u, v) + A_j(u, v) - g_i(u) - g_j(v)]. \quad (4.7)$$

Indeed, (4.2) with $\lambda = 2$ converts the reference interval integral and its endpoint factors into $B_0(z)$. The actual separation is z if $i = j$, $D + z$ if $(i, j) = (0, 1)$, and $D - z$ if $(i, j) = (1, 0)$. In the last case $x > y$; the bound in Proposition 3.5 includes this order. All three separations have the same absolute fast model factor $B_0(z)$.

The four weights $w_{ij}(z) = w(|z + (j - i)D|)$ form the matrix

$$(w_{ij}(z))_{i,j=0}^1 = B_0(z)^k \begin{pmatrix} \sin^m z & \cos^m z \\ \cos^m z & \sin^m z \end{pmatrix}. \quad (4.8)$$

Every row and every column sums to

$$W_D(z) = (\sin^m z + \cos^m z) B_0(z)^k.$$

This is symmetric under $z \mapsto D - z$, and

$$\frac{W_D(z)}{\sin(2z)} = 2^{-k} (\sin^m z + \cos^m z) \sin^{k-1}(2z)$$

is bounded for every $k \geq 1$. When $k = 1$, its limit at either endpoint is $1/2$, so this case also satisfies Lemma 4.1.

Multiply (4.7) by $w_{ij}(z)$ and sum over i, j . The row sums control the terms indexed by i , and the column sums control those indexed by j , giving exactly

$$\sum_{i,j=0}^1 w_{ij}(z) \frac{L_K(u + iD, v + jD)}{k} \geq W_D(z) \sum_{i=0}^1 \left[A_i(u, v) - \frac{g_i(u) + g_i(v)}{2} \right]. \quad (4.9)$$

Lemma 4.1, applied on $(0, D)$ separately to s_0 and s_1 with weight W_D , shows that the right side has nonnegative integral over Δ_D .

For completeness, the four images of Δ_D used above consist precisely of pairs whose first endpoint has smaller residue modulo D than the second. Their transposes contain precisely the opposite residue order. These eight disjoint regions cover $\mathcal{R} \setminus E$; each coordinate change has Jacobian one. Since $w_{xy} L_K(x, y)$ is symmetric, it follows that

$$\begin{aligned} & \int_{\mathcal{R}} w_{xy} L_K(x, y) dx dy \\ &= 2k \int_{\Delta_D} \sum_{i,j=0}^1 w_{ij}(v - u) \frac{L_K(u + iD, v + jD)}{k} du dv \geq 0. \end{aligned} \quad (4.10)$$

Here the factor 2 accounts for endpoint exchange and the factor k restores the logarithm of f_K from that of its k th root.

Comparison and equality. Adding (4.6) and (4.10) gives

$$\int_{\mathcal{R}} w_{xy} \log(f/w_{xy}) dx dy \geq 0.$$

For $r > 0$, the strictly convex function

$$\Phi(r) = r - 1 - \log r$$

is nonnegative and vanishes exactly at $r = 1$. On $\mathcal{R} \setminus E$ put $r = f/w_{xy}$. All terms in the identity

$$\int_{\mathcal{R}} (f - w_{xy}) dx dy = \int_{\mathcal{R}} w_{xy} \log r dx dy + \int_{\mathcal{R}} w_{xy} \Phi(r) dx dy \quad (4.11)$$

are finite by the preceding integrability argument. Both terms on the right are nonnegative, proving (4.4). If equality holds, the last integral is zero. Since $w_{xy} > 0$ off the diagonal and cut lines, $\Phi(r) = 0$ almost everywhere, hence $f = w_{xy}$ almost everywhere in $\mathcal{R}$. Continuity of both functions on $[0, T]^2$ extends this identity to every pair, including the exceptional lines and boundary. Conversely, that pointwise identity gives equality in (4.4). $\square$

5 Global comparison and the equality case

We retain all intermediate model types and the normalization of Section 2. The preceding geodesic comparison now gives the global volume bound. Its nonnegative defect will also retain enough information to identify the equality metric.

For $\xi \in U$, write γ_ξ for the geodesic with initial velocity ξ and f_ξ for the determinant of Proposition 3.1. Put

$$J(\xi, t) = f_\xi(0, t).$$

This is the absolute determinant of the normal Jacobi tensor with initial value zero and derivative Id. It is continuous in (ξ, t) , positive for $0 < t < D$, and agrees there with the usual polar volume density.

Lemma 5.1 (Polar integration over a full period). *For every $p \in M$,*

$$\int_{S_p M} \int_0^D J(\xi, t) dt d\sigma_p(\xi) = V, \quad \int_{S_p M} \int_0^T J(\xi, t) dt d\sigma_p(\xi) = 2V.$$

Proof. The open injectivity ball maps diffeomorphically onto the complement of the cut locus. The latter is a smooth submanifold of dimension $d - k < n$, hence has zero Riemannian volume. Polar coordinates therefore prove the first formula. Periodicity of the Jacobi data and reversal of the geodesic give

$$J(\xi, T - t) = J(\xi, -t) = J(-\xi, t).$$

The sign in the reversed initial derivative disappears under the absolute determinant. The substitution $t \mapsto T - t$ in the second half-period, followed by the area-preserving antipodal map on $S_p M$, proves the second formula. $\square$

Proof of Theorem 1.2. The circle, sphere-type, and real-projective-type cases are already covered in Section 2. In an intermediate case use the normalization (2.1). Theorem 4.2 says that

$$\delta(\xi) := \int_0^T \int_0^T (f_\xi(x, y) - w(|y - x|)) dx dy \geq 0. \quad (5.1)$$

Time translation along a geodesic gives

$$f_\xi(x, y) = J(\phi_x \xi, (y - x) \bmod T).$$

The possible choice of endpoint for the residue modulo T has no effect on any integral. Fubini's theorem, translation on the time circle, flow invariance of μ , and Lemma 5.1 yield

$$\begin{aligned} \int_U \int_0^T \int_0^T f_\xi(x, y) dx dy d\mu(\xi) &= T \int_U \int_0^T J(\xi, t) dt d\mu(\xi) \\ &= 2TV^2. \end{aligned}$$

All integrands are continuous and bounded on the compact domains in question. Since $w(T-t) = w(t)$, its periodic extension gives likewise

$$\begin{aligned} \int_U \int_0^T \int_0^T w(|y-x|) dx dy d\mu(\xi) &= \sigma_d VT \int_0^T w(t) dt \\ &= 2TVV_0. \end{aligned}$$

Integrating (5.1) and dividing by $2TV > 0$ proves $V \geq V_0$.

If $V = V_0$, the nonnegative function δ has integral zero. It is continuous, because the underlying Jacobi tensors depend smoothly on the initial velocity and taking the absolute determinant preserves continuity. Full support of μ forces $\delta(\xi) = 0$ for every ξ . The equality assertion in Theorem 4.2 now gives

$$J(\xi, t) = w(t), \quad \xi \in U, \quad 0 < t < D. \quad (5.2)$$

In Cartesian normal coordinates at any point, the volume density is $J(\xi, t)/t^d$. By (5.2) it depends only on t . At the origin it extends smoothly with value one, as also follows directly from the normal-coordinate volume form. Thus M is harmonic. It is compact and simply connected by Proposition 2.2. The compact simply connected case of the Lichnerowicz conjecture, proved by Szabó [25], identifies its metric with a compact rank-one symmetric metric up to scale. Its type identifies the model, and the fixed diameter identifies the scale. For the radial-density definition and the concluding metric classification, see also the author's original preprint [24, pp. 2–4 and Lemmas 4.9–4.10].

Conversely, a model is isometric to itself and has volume V_0 . Finally both sides of the volume comparison scale by the same factor under a constant metric rescaling. Undoing (2.1) proves the assertion at every diameter. $\square$

Proof of Corollary 1.3. The upper bound and Theorem 1.2 give equal volumes, so its equality case applies. $\square$

6 Incidence and the quaternionic Pontryagin class

The lower-volume comparison and its metric equality case are now proved. To obtain the quaternionic upper bound, we first determine the characteristic class of an actual bundle supplied by the cut geometry: the normal four-plane bundle of cut incidence. The later loop-energy construction will use this same bundle, not only its sphere-fibration homotopy type. We use the standard integral bundle and cohomology tools in Hatcher [9] and Milnor and Stasheff [16]; the geometric constructions and the relations needed here are proved below. In this section and the next three, suppose that M has quaternionic type, so $n = 4r$, $r \geq 2$, and $k = 3$. Choose an integral generator $u \in H^4(M; \mathbb{Z})$ and an orientation with $\int_M u^r = 1$. Write $C_p = \text{Cut}(p)$. The cell attachment in Section 2 gives

$$H^*(C_p; \mathbb{Z}) = \mathbb{Z}[u|_{C_p}]/((u|_{C_p})^r).$$

Indeed restriction is an isomorphism in every degree below $4r - 1$, and C_p has dimension $4r - 4$. In particular it is orientable, and its fundamental class is dual to $\pm u$ in M .

6.1 The actual normal bundle

Proposition 6.1. *The cut incidence*

$$F = \{(p, q) \in M \times M : q \in C_p\}$$

is a smooth embedded submanifold. Both projections are smooth fiber bundles with fiber a cut locus. If u_1, u_2 are the two pullbacks of u , its rank-four normal bundle $\mathcal{V}$ can be oriented so that

$$e(\mathcal{V}) = u_1 - u_2, \quad p_1(\mathcal{V}) = 2s(u_1 + u_2), \quad s \in 2\mathbb{Z} + 1. \quad (6.1)$$

Projection of $\mathcal{V}_{(p,q)}$ into $T_p M$ identifies it with $N_p C_q$. The unit sphere of this four-plane is exactly the great sphere of initial directions of minimizing geodesics from p to q .

Proof. The full cut map

$$c : SM \longrightarrow M \times M, \quad c(\xi) = (\pi(\xi), \exp_{\pi(\xi)}(D\xi))$$

has constant rank $4r + (4r - 4) = 8r - 4$. Its fibers are the connected great three-spheres from Proposition 2.2. Here constant rank and compactness give an embedded image, as follows. Constant-rank charts give local image germs through any image point. Charts overlapping along its fiber have the same germ, so connectedness of that fiber makes all these germs agree. Cover the compact fiber by finitely many such charts. Properness excludes additional inverse images outside their union over a sufficiently small neighborhood of the image point. Thus the whole image in that neighborhood is the one smooth submanifold germ. This proves the assertion. The first projection is a submersion because its composition with c is π ; symmetry gives the same conclusion for the second. Proper submersions are smooth fiber bundles.

For $(a, b) \in \mathcal{V}_{(p,q)} \subset T_p M \oplus T_q M$, orthogonality to $(T_p C_q, 0)$ gives $a \in N_p C_q$. If $a = 0$, surjectivity of the second projection on $T_{(p,q)} F$ forces $b = 0$. The projection to $N_p C_q$ is therefore injective and, both spaces having dimension four, an isomorphism. First variation for the constant distance D shows that the initial directions from p to q are normal to C_q . Their great sphere has the full dimension of $S(N_p C_q)$, proving the last assertion. In particular, if $q_1, q_2 \in C_p$ are distinct, then

$$N_p C_{q_1} \cap N_p C_{q_2} = \{0\} : \quad (6.2)$$

a common nonzero direction would reach both points at time D .

The cut loci and M are simply connected, so the fibration makes F simply connected. Its normal bundle is orientable. Leray–Hirsch [9, Theorem 4D.1], using $1, u_2, \dots, u_2^{r-1}$ on the fibers, and disjointness from the diagonal give the integral presentation

$$H^*(F; \mathbb{Z}) = \mathbb{Z}[u_1, u_2] / (u_1^{r+1}, \Delta(u_1, u_2)), \quad \Delta(a, b) = \sum_{j=0}^r a^j b^{r-j}. \quad (6.3)$$

To justify completeness, the displayed relation is the restriction of the diagonal class in $M \times M$. It is monic of degree r in u_2 , so the proposed quotient has exactly the Leray–Hirsch basis. The dual of F in $M \times M$ restricts on either factor slice to $\pm u$. Its restriction to the disjoint diagonal is zero. Its two factor coefficients are consequently opposite, and after an orientation choice its restriction to F is $u_1 - u_2$. The self-intersection formula gives the asserted Euler class.

The interchange of factors preserves the unoriented normal bundle. Since u_1, u_2 are an integral basis of $H^4(F; \mathbb{Z})$, we have $p_1(\mathcal{V}) = t(u_1 + u_2)$ for an integer t . Also $H^2(F; \mathbb{Z}/2) = 0$, so $\mathcal{V}$ is spin. We use the classical identification $\text{Spin}(4) = \text{SU}(2) \times \text{SU}(2)$ and its two complex two-dimensional half-spin representations; see Moore [17, Sections 2.1–2.2]. For an oriented spin four-plane bundle, its two complex half-spin bundles have second Chern classes

$$\frac{-p_1(\mathcal{V}) + 2e(\mathcal{V})}{4}, \quad \frac{-p_1(\mathcal{V}) - 2e(\mathcal{V})}{4}.$$

This follows directly from the formal half-spin weights $\pm(t_1 + t_2)/2$ and $\pm(t_1 - t_2)/2$, where $p_1 = t_1^2 + t_2^2$ and $e = t_1 t_2$. Integrality in the basis u_1, u_2 implies $t \equiv 2 \pmod{4}$. Writing $t = 2s$ proves the proposition. $\square$

Changing u to $-u$ replaces s by $-s$. Reverse the orientation of $\mathcal{V}$ at the same time to retain $e(\mathcal{V}) = u_1 - u_2$, and reverse the orientation of M exactly when r is odd to retain $\int_M u^r = 1$. We may therefore take $s \geq 1$.

6.2 The rank constraint from two direction planes

For distinct $q_1, q_2 \in C_p$, (6.2) places two four-planes with zero intersection inside $T_p M$. Their direct sum has a complement of rank $4(r - 2)$, which limits the degree of its total Pontryagin class. We will use this rank restriction to determine $p(TM)$ from s . The parameter space must therefore exclude coincident cut points; put

$$Z = \{(p, q_1, q_2) : q_1, q_2 \in C_p, q_1 \neq q_2\}.$$

Use v, x_1, x_2 for the pullbacks of u from its three points. In this section polynomial degree is one quarter of cohomological degree. In degrees divisible by four the rational cohomology of Z is the algebra

$$A = \mathbb{Q}[v, x_1, x_2]/(v^{r+1}, \Delta(v, x_1), S(v, x_1, x_2)), \quad S(v, x_1, x_2) = \sum_{i+j+\ell=r-1} v^i x_1^j x_2^\ell. \quad (6.4)$$

Here and below this assertion identifies the subalgebra consisting of these degrees; no claim about other degrees is needed.

For completeness, before removing $q_1 = q_2$, Leray–Hirsch gives the relations v^{r+1} , $\Delta(v, x_1)$ and $\Delta(v, x_2)$. The relative diagonal has codimension $4(r - 1)$. Its dual is $\pm S$. To see this, the dual is a homogeneous polynomial of degree $r - 1$ annihilated by $x_1 - x_2$. In degree r the only relations are the two displayed Δ polynomials; their difference is $(x_1 - x_2)S$. More explicitly, multiplying a representative of the dual by $x_1 - x_2$ gives a constant linear combination of those two relations. Setting $x_1 = x_2$ makes the two constants opposite. Cancellation of $x_1 - x_2$ then takes place in the ordinary polynomial ring, before passage to cohomology, and shows that the dual is proportional to S . Restriction to a fixed- p fiber gives the ordinary diagonal class of $C_p \times C_p$, fixing the factor as ± 1 . The Thom sequence for removing the relative diagonal is surjective onto the complement in degrees divisible by four, because the relevant next cohomology group of F is zero. Restriction from the ambient fiber product to that diagonal is surjective, so the projection formula identifies the kernel with the ideal of its dual. The relation $\Delta(v, x_2)$ then becomes redundant, proving (6.4). Successive division by its two monic relations proves that A is free over $\mathbb{Q}[v]/(v^{r+1})$, with basis

$$\{x_1^a x_2^b : 0 \leq a < r, 0 \leq b < r - 1\}. \quad (6.5)$$

Theorem 6.2. *The total Pontryagin class of M is*

$$p(TM) = \frac{(1 + 2su + u^2)^{r+1}}{1 + 4su} \quad \text{in } H^*(M; \mathbb{Q}). \quad (6.6)$$

The quotient means its power-series expansion truncated by $u^{r+1} = 0$.

Proof. Over Z , the two actual normal four-planes in (6.2) have direct sum inside the pullback of TM from p . Choose a complementary bundle W of rank $4(r - 2)$. Set

$$D_s(v, x) = 1 + 2s(v + x) + (v - x)^2, \quad p_0(v) = \frac{(1 + 2sv + v^2)^{r+1}}{1 + 4sv}.$$

The rank-four identity $p_2 = e^2$ [16, Corollary 15.8] and Proposition 6.1 give

$$p(W) = \frac{p(TM)(v)}{D_s(v, x_1)D_s(v, x_2)}. \quad (6.7)$$

Its polynomial degree is at most $t_0 = 2r - 4$.

We will prove that the formal characteristic class

$$\mathcal{W}_0 = \frac{p_0(v)}{D_s(v, x_1)D_s(v, x_2)} \in A$$

has degree at most t_0 and a degree- t_0 component that is nonzero modulo v . The freeness in (6.5) will then show that any first discrepancy between $p(TM)(v)$ and $p_0(v)$ violates the rank bound on W . For the degree bound we use only the algebra (6.4), without presuming the answer for TM . Successive monic polynomial divisions factor

$$T^{r+1} = (T - v)(T - x_1)(T - x_2) \prod_{j=4}^{r+1} (T - z_j)$$

in an extension of A obtained by adjoining roots of monic polynomials. Give each root degree one; the monic relations are homogeneous for this grading. Each adjoining algebra is free with a basis containing 1, so this extension is injective. The positive-degree elementary symmetric functions of all $r + 1$ roots vanish. Consequently, the product of $D_s(v, z)$ over all these roots is $(1 + 2sv + v^2)^{r+1}$. The factor at $z = v$ is $1 + 4sv$. Dividing these constant-one series therefore gives

$$\mathcal{W}_0 = \prod_{j=4}^{r+1} D_s(v, z_j).$$

There are $r - 2$ factors, each of degree at most two, as required.

We also need the top component $(\mathcal{W}_0)_{t_0}$ to be nonzero modulo v . For $r = 2$ it is 1. For $r \geq 3$, let

$$(1 + 2sx + x^2)^{-1} = \sum_{j \geq 0} a_j x^j.$$

The recursion $a_0 = 1$, $a_1 = -2s$ and $a_j = -2sa_{j-1} - a_{j-2}$ gives

$$a_j^2 - a_{j-1}a_{j+1} = 1 \quad (j \geq 1). \quad (6.8)$$

Indeed the left side is unchanged on replacing j by $j - 1$, and its initial value is one. Modulo v the algebra can also be presented as

$$\mathbb{Q}[x_1, x_2] / \left(x_1^r, x_2^r, \sum_{j=0}^{r-1} x_1^j x_2^{r-1-j} \right).$$

In degree $2r - 4$ the surviving monomials after the first two truncations are

$$x_1^{r-3}x_2^{r-1}, \quad x_1^{r-2}x_2^{r-2}, \quad x_1^{r-1}x_2^{r-3}.$$

The remaining ideal gives exactly their sum as a relation: multiplying its generator by any monomial of degree $r - 3$ gives this same sum. Their coefficients in $\mathcal{W}_0$ are, in order, $a_{r-3}a_{r-1}$, a_{r-2}^2 , $a_{r-3}a_{r-1}$. Equation (6.8) makes the middle coefficient differ from either outside coefficient by one. Thus the component modulo v is the nonzero class of $x_1^{r-2}x_2^{r-2}$, one of the monomials in (6.5).

Finally suppose that $p(TM)(v)/p_0(v)$ first differs from 1 in degree i , where $1 \leq i \leq r$. Write its first coefficient as $b \neq 0$. In degree $t_0 + i$, modulo v^{i+1} , (6.7) is $bv^i(\mathcal{W}_0)_{t_0}$. This is nonzero by (6.5) and the preceding calculation: in a free module over $\mathbb{Q}[v]/(v^{r+1})$, a coefficient nonzero modulo v remains nonzero after multiplication by v^i , modulo v^{i+1} , for $i \leq r$. It contradicts the rank bound on W . Therefore there is no such i , which proves the theorem. $\square$

7 The signature determines the incidence coefficient

The incidence construction has reduced the bundle calculation to one odd integer s . The signature theorem determines it in every quaternionic rank. The small ranks follow from exact coefficients; a single contour estimate treats all larger ranks.

Theorem 7.1. *In Proposition 6.1, $|s| = 1$. In particular, the actual tautological four-plane bundle of every tangent great-sphere fibration at a point of M satisfies $p_1 = \pm 2e$.*

Proof. Keep the choices $s \geq 1$ and $\int_M u^r = 1$. Set

$$a_+ a_- = 1, \quad a_+^2 + a_-^2 = 2s, \quad a_+ \geq a_- > 0, \quad c = 2\sqrt{s}, \quad f(y) = \frac{\sqrt{y}}{\tanh \sqrt{y}}.$$

Hirzebruch's signature theorem [16, Signature Theorem 19.4], applied to Theorem 6.2 gives

$$\sigma_r = [u^r] \frac{(f(a_+^2 u) f(a_-^2 u))^{r+1}}{f(c^2 u)}. \quad (7.1)$$

Here f is its power series at zero; the two factors reflect $1 + 2su + u^2 = (1 + a_+^2 u)(1 + a_-^2 u)$. The integral cohomology ring of M gives

$$\sigma_r = \begin{cases} 1, & r \text{ even,} \\ 0, & r \text{ odd.} \end{cases}$$

Indeed for even r the middle pairing has the positive basis $u^{r/2}$, and for odd r its middle cohomology is zero. Expanding $f(y) = 1 + y/3 - y^2/45 + 2y^3/945 + \dots$ in (7.1) gives

$$\sigma_2 - 1 = \frac{8(s^2 - 1)}{15}, \quad \sigma_3 = -\frac{32s(s^2 - 1)}{105}.$$

These prove the assertion for $r = 2, 3$.

We give uniform estimates for all $r \geq 4$. Since s is odd, the contrary case is $s \geq 3$. We express the signature as a residue and show that the two nearest nonzero poles give a main term that dominates the remaining contour integral, forcing $(-1)^r \sigma_r > 1$. Put

$$\tau = \frac{a_-}{a_+} = s - \sqrt{s^2 - 1}, \quad b = \frac{c}{a_+} = \sqrt{2(1 + \tau^2)}, \quad Q(z) = \tan z \tan(\tau z), \quad \rho = \frac{\pi}{2b}.$$

Then $0 < \tau \leq 3 - \sqrt{8} < 0.172$ and $\sqrt{2} < b < 1.435$. Substitute $u = -t^2$ in (7.1). Since $f(-a^2 t^2) = at / \tan(at)$ and $a_+ a_- = 1$, taking the coefficient by a residue and then putting $z = a_+ t$ gives

$$(-1)^r \sigma_r = \frac{\tau}{b} \operatorname{Res}_{z=0} \frac{\tan(bz)}{Q(z)^{r+1}} dz. \quad (7.2)$$

The change of variables contributes the factor $1/(ca_+) = \tau/b$; this fixes the normalization of the residue.

Contour representation. Take the rectangle with vertical sides $\Re z = \pm X$, $X = 3\pi/4$, and horizontal sides $\Im z = \pm 4$. Apart from zero, the only poles of the integrand in this rectangle are at $z = \pm \rho$. Zeros of $\tan z$ occur only at multiples of π , zeros of $\tan(\tau z)$ only at multiples of π/τ , and the next poles of $\tan(bz)$ are outside because $3\pi/(2b) > X$. The poles of $\tan z$ at $\pm \pi/2$ give zeros of the reciprocal denominator, not poles. Both residues at $\pm \rho$ equal $-b^{-1} Q(\rho)^{-r-1}$, with $Q(\rho) > 0$. The residue theorem therefore expresses (7.2) as

$$(-1)^r \sigma_r = L + \mathcal{E}, \quad L = \frac{2\tau}{b^2} Q(\rho)^{-r-1}. \quad (7.3)$$

Symmetry of the four sides and the triangle inequality bound $|\mathcal{E}|/L$ by the sum of

$$E_v = \frac{b}{\pi} \int_0^4 |\tan(b(X + iy))| \left(\frac{Q(\rho)}{|Q(X + iy)|} \right)^{r+1} dy, \quad (7.4)$$

$$E_h = \frac{b}{\pi} \int_0^X |\tan(b(x + 4i))| \left(\frac{Q(\rho)}{|Q(x + 4i)|} \right)^{r+1} dx. \quad (7.5)$$

Uniform bounds. The same contour works for every $r \geq 4$ once both boundary ratios are bounded by constants below one: their fifth powers then control every exponent $r + 1 \geq 5$. We retain the τ dependence in the numerator and denominator so these bounds remain uniform as $\tau \downarrow 0$. All finite decimals below denote the exact indicated rational numbers. Using $3.14 < \pi < 3.142$ gives

$$\rho < 1.111, \quad \tan \rho < 2.022, \quad \tau \rho < 0.192, \quad Q(\rho) < 2.30\tau.$$

For the last inequality, the increase of $\tan t/t$ and the Taylor bounds on $[0, 0.192]$ give $\tan t/t < 1.013$, and $2.022 \cdot 1.111 \cdot 1.013 < 2.30$. The same monotonicity gives

$$\frac{Q(\rho)}{\tan(\tau X)} \leq \frac{\rho \tan \rho}{X} < 0.955. \quad (7.6)$$

On the vertical sides, $|\tan(X + iy)| = 1$. Also $\cos(2bX) \geq 0$, since $2bX = 3b\pi/2 \in (2\pi, 5\pi/2)$, and hence $|\tan(b(X + iy))| \leq 1$. The elementary complex tangent identity gives

$$\begin{aligned} \frac{|\tan(\tau(X + iy))|^2}{\tan^2(\tau X)} &= \frac{1 + \sinh^2(\tau y)/\sin^2(\tau X)}{1 + \sinh^2(\tau y)/\cos^2(\tau X)} \\ &\geq 1 + 0.08y^2 \quad (0 \leq y \leq 4). \end{aligned}$$

For this last bound, the numerator is at least $1 + y^2/X^2$, whereas $\cos(\tau X) > 0.917$ and $\sinh(\tau y) \leq 1.082\tau y$ bound the denominator by $1 + Ay^2$ with $A = (1.082 \cdot 0.172/0.917)^2$. The inequality follows from $X^{-2} \geq A + 0.08 + 16(0.08)A$. Since $r + 1 \geq 5$, (7.6) now implies

$$\begin{aligned} E_v &< \frac{1.435}{3.14} (0.955)^5 \int_0^4 (1 + 0.08y^2)^{-5/2} dy \\ &< 0.783. \end{aligned}$$

Here the integral equals

$$\frac{4(1 + 2 \cdot 0.08 \cdot 16/3)}{(1 + 0.08 \cdot 16)^{3/2}} < 2.154.$$

On a horizontal side the identity for the modulus of tangent gives

$$|Q(x + 4i)| \geq \tanh 4 \tanh(4\tau), \quad |\tan(b(x + 4i))| \leq \coth(4b) < 1.001.$$

Since $\tau/\tanh(4\tau)$ is increasing,

$$\frac{Q(\rho)}{\tanh 4 \tanh(4\tau)} < \frac{2.30 \cdot 0.172}{0.999 \cdot 0.595}.$$

Substitution in (7.5), with $X/\pi = 3/4$, therefore gives

$$E_h < \frac{3 \cdot 1.435}{4} 1.001 \left(\frac{2.30 \cdot 0.172}{0.999 \cdot 0.595} \right)^5 < 0.142.$$

The numerical trigonometric inequalities used here can be checked without numerical approximation: use

$$\sin t \leq t - t^3/6 + t^5/120, \quad \cos t \geq 1 - t^2/2 + t^4/24 - t^6/720 \quad (0 \leq t \leq 1.111)$$

for $\tan \rho$ and the smaller real arguments. For the hyperbolic inequalities use

$$\frac{\sinh t}{t} \leq 1 + \frac{t^2}{6} + \frac{t^4/120}{1 - t^2/42} \quad (0 \leq t \leq 0.688).$$

The first seven terms of the exponential series at 1.376 give $\tanh(0.688) > 0.595$, while $e^8 > 2000$ gives $\tanh 4 > 0.999$. The same bound and $b > \sqrt{2}$ imply $\coth(4b) < 1.001$.

Finally $Q(\rho) < 2.30\tau < 1$, so (7.3) and $r + 1 \geq 5$ imply

$$L > \frac{\tau}{1 + \tau^2} (2.30\tau)^{-5} \geq \frac{1}{(1 + 0.172^2) 2.30^5 0.172^4} > 17.$$

Combining these estimates yields

$$(-1)^r \sigma_r > (1 - 0.783 - 0.142)17 = 1.275 > 1,$$

contradicting either possible cohomological signature. Thus $s = 1$. Restricting (6.1) to a fixed- p fiber gives $e(\mathcal{V}) = -u_2$ and $p_1(\mathcal{V}) = 2u_2$ there. Proposition 6.1 identifies this restriction with the actual tautological four-plane bundle, which proves the last assertion as well. $\square$

8 The geodesic space and its integral classes

Throughout this section, M has quaternionic model type, so that

$$n = 4r, \quad r \geq 2, \quad d = 4r - 1, \quad k = 3, \quad H^*(M; \mathbb{Z}) = \mathbb{Z}[u]/(u^{r+1}), \quad |u| = 4.$$

We retain the normalization $D = \pi/2$ and $T = 2D = \pi$. Choose the sign of u as permitted by Theorem 7.1, so that the actual four-plane bundle $\mathcal{V}$ on cut incidence F satisfies

$$p_1(\mathcal{V}) = 2(u_1 + u_2). \tag{8.1}$$

Orient M by $\int_M u^r = 1$. Only the characteristic-class conclusion (8.1) and the bundle identification in Proposition 6.1 are used from the incidence argument.

Let $U = SM$, let $\pi : U \rightarrow M$ be the unit tangent projection, and write $z = \pi^*u$. The prime period is T , so the action

$$(\mathbb{R}/\mathbb{Z}) \times U \longrightarrow U, \quad (a, w) \longmapsto \phi_{Ta}(w)$$

is free. Its quotient and quotient map will be denoted by

$$B = U/S^1, \quad \rho : U \longrightarrow B.$$

Thus B is the closed smooth manifold of oriented, unmarked geodesics, of dimension $2n - 2$.

8.1 The contact normalization and the two Gysin sequences

The contact calculation is the volume method for periodic geodesic flows associated with Weinstein [26]; we include it to fix the period, Euler sign and angular-area factors exactly. The ensuing integral Gysin sequences, universal coefficient theorem and Poincaré duality are used in their standard forms [9, 16]. Yang's complex-projective volume calculation also uses the Gysin sequences of the unit-tangent sphere bundle and the geodesic-circle bundle [28, pp. 381–382]. Here the degree-four generator leads to a different middle-degree lattice.

Let λ be the unit tangent contact form,

$$\lambda_w(\zeta) = \langle w, d\pi_w \zeta \rangle.$$

Its Reeb flow is the unit-speed geodesic flow. Consequently $\theta = \lambda/T$ is a connection for the above $\mathbb{R}/\mathbb{Z}$ action: it is invariant and integrates to one on each orbit. Its curvature is the pullback of a symplectic form ω_B on B . We fix the Euler-class sign convention by

$$\rho^* \omega_B = d\theta, \quad e = [\omega_B] \in H^2(B; \mathbb{Z}),$$

and give B its symplectic orientation.

Proposition 8.1 (Contact volume). *With this orientation and Euler class,*

$$N := \int_B e^{4r-1} = \frac{(4r-1)! |S^{4r-1}| \text{Vol}(M)}{T^{4r}} > 0. \quad (8.2)$$

For the standard quaternionic model at the same diameter, this number is

$$N_0 = \frac{1}{2r+1} \binom{4r}{2r}. \quad (8.3)$$

Proof. Integration over the circle fibers gives

$$\int_B e^d = \int_U \theta \wedge (d\theta)^d = T^{-(d+1)} \int_U \lambda \wedge (d\lambda)^d,$$

where U has the orientation induced by the positive circle direction and the chosen orientation of B . To compute the last form as a density, use the connection splitting of $T_w U$ into horizontal and vertical directions. Choose an orthonormal frame $(w, e_1, \dots, e_d)$ at $\pi(w)$. The contact form reads off the horizontal w coordinate, and $d\lambda$ pairs each horizontal e_i coordinate with its vertical e_i coordinate, up to one common sign convention. In the d -fold exterior power each pairing occurs once in each of $d!$ orders. Hence

$$|\lambda \wedge (d\lambda)^d| = d! d\mu,$$

where $d\mu$ is base Riemannian volume times unit tangent sphere area. The chosen orientation makes the contact integral positive, and $\mu(U) = |S^d| \text{Vol}(M)$. Since $d+1 = 4r$, this proves (8.2).

The model's polar density on $(0, D)$ is $\sin^{4r-1} t \cos^3 t$. With $y = \sin^2 t$,

$$\int_0^{\pi/2} \sin^{4r-1} t \cos^3 t dt = \frac{1}{2} \int_0^1 y^{2r-1} (1-y) dy = \frac{1}{4r(2r+1)}.$$

Using $|S^{4r-1}| = 2\pi^{2r}/(2r-1)!$ and $T = \pi$ in (8.2) gives

$$N_0 = \frac{(4r-1)!}{r(2r+1)((2r-1)!)^2} = \frac{(4r)!}{(2r)!(2r+1)!},$$

which is (8.3). □

Thus the desired upper volume bound is $N \leq N_0$. The integral cohomology calculation below computes the determinant of a middle pairing whose entries are top-degree products, including this characteristic number.

Lemma 8.2 (Integral Gysin information). *The manifolds U and B are simply connected. There is an integral class $v \in H^4(B; \mathbb{Z})$ with $\rho^* v = z$. For $2j < n$, the following is an integral basis:*

$$H^{2j}(B; \mathbb{Z}) : \quad e^{j-2a} v^a, \quad 0 \leq a \leq \lfloor j/2 \rfloor. \quad (8.4)$$

All odd cohomology of B vanishes, and its even cohomology is free abelian. In particular $H^2(B; \mathbb{Z}) = \mathbb{Z}e$ and $H^4(B; \mathbb{Z}) = \mathbb{Z}e^2 \oplus \mathbb{Z}v$. The rational cohomology ring is generated by e, v , and

$$\dim_{\mathbb{Q}} H^*(B; \mathbb{Q}) = 2r(r+1). \quad (8.5)$$

Multiplication by e at the middle fits into an exact sequence

$$0 \longrightarrow H^{n-2}(B; \mathbb{Z}) \xrightarrow{e \smile} H^n(B; \mathbb{Z}) \longrightarrow \mathbb{Z}/(r+1) \longrightarrow 0. \quad (8.6)$$

Consequently the integral bilinear form

$$(a, b) \mapsto \int_B eab, \quad a, b \in H^{n-2}(B; \mathbb{Z}), \quad (8.7)$$

has determinant of absolute value $r+1$ in any integral basis. The map $e \smile: H^n(B; \mathbb{Z}) \rightarrow H^{n+2}(B; \mathbb{Z})$ is an isomorphism.

Proof. The sphere fiber of π has dimension $n-1 \geq 7$. Its homotopy sequence and simple connectivity of M give $\pi_1(U) = 0$. The homotopy sequence of the circle bundle then gives $\pi_1(B) = 0$.

The Euler number of TM is $\chi(M) = r+1$. The chosen orientation therefore gives

$$e(TM) = (r+1)u^r.$$

The integral Gysin sequence of the oriented sphere bundle π has the form

$$H^{q-n}(M) \xrightarrow{e(TM) \smile} H^q(M) \xrightarrow{\pi^*} H^q(U) \xrightarrow{\pi_!} H^{q-n+1}(M) \xrightarrow{e(TM) \smile} H^{q+1}(M).$$

It gives the complete additive calculation

$$H^q(U; \mathbb{Z}) = \begin{cases} \mathbb{Z}\{z^a\}, & q = 4a, \quad 0 \leq a < r, \\ \mathbb{Z}/(r+1)\{z^r\}, & q = n, \\ \mathbb{Z}\{y_a\}, & q = n-1+4a, \quad 1 \leq a \leq r, \\ 0, & \text{otherwise.} \end{cases} \quad (8.8)$$

Here the upper generators may be chosen with $\pi_! y_a = u^a$ and $y_a = z^{a-1} y_1$; the latter assertion follows from the projection formula and the fact that $\pi_!$ is an isomorphism in these upper degrees. At degree $n-1$, the Gysin map from $H^0(M)$ to $H^n(M)$ is multiplication by the nonzero integer $r+1$, so $H^{n-1}(U) = 0$. This also explains the sole torsion group in (8.8).

Apply next the integral Gysin sequence of ρ . The vanishing of odd groups of U below degree n inductively gives $H^{2j-1}(B) = 0$ for $2j-1 < n$. For positive even $2j < n$ the remaining terms give

$$0 \longrightarrow H^{2j-2}(B) \xrightarrow{e \smile} H^{2j}(B) \xrightarrow{\rho^*} H^{2j}(U) \longrightarrow 0. \quad (8.9)$$

For $2j=2$ this says $H^2(B) = \mathbb{Z}e$. For $2j=4$, the last group is freely generated by z , so choose an integral lift v . If j is odd, the last group in (8.9) vanishes and multiplication by e gives the next basis. If $j=2a$ is even, the last group is generated by z^a , and the class v^a lifts that generator. Induction proves exactly the integral basis (8.4).

In all degrees at most $n-2$, the homology groups of B are free and have the same ranks as its cohomology groups. Indeed the universal coefficient theorem detects torsion in $H_i(B; \mathbb{Z})$ as torsion in $H^{i+1}(B; \mathbb{Z})$, and the latter groups are free through degree $n-1$. Poincaré duality on the oriented $(2n-2)$ -manifold B now supplies all its cohomology in degrees at least n . It is free in even degrees and zero in odd degrees; together with the low-degree calculation this proves the asserted global statement.

The circle Gysin sequence at degree n uses $H^{n-1}(U) = H^{n-1}(B) = 0$ and $H^n(U) = \mathbb{Z}/(r+1)$, and is precisely (8.6). The Poincaré pairing

$$H^{n-2}(B; \mathbb{Z}) \times H^n(B; \mathbb{Z}) \longrightarrow \mathbb{Z}$$

is unimodular, since these two groups are free. Composing one factor with the injection of index $r + 1$ in (8.6) proves the determinant assertion (8.7). Finally $H^{n+1}(U) = H^{n+2}(U) = 0$ by (8.8). The next portion of the circle Gysin sequence therefore makes multiplication by e from degree n to degree $n + 2$ an isomorphism.

In every even degree $q \geq n$, the group $H^q(U; \mathbb{Q})$ vanishes. The rational circle Gysin sequence therefore makes $e \smile: H^{q-2}(B; \mathbb{Q}) \rightarrow H^q(B; \mathbb{Q})$ surjective. The low-degree bases now show that the full rational ring is generated by e, v . Below degree n , the basis ranks sum to

$$\sum_{j=0}^{2r-1} (\lfloor j/2 \rfloor + 1) = 2 \sum_{a=0}^{r-1} (a + 1) = r(r + 1).$$

Poincaré duality supplies another $r(r + 1)$ dimensions above degree $n - 1$, and the middle odd group is zero. This proves (8.5). $\square$

8.2 The first free-loop attachment

The Gysin sequences determine the low-degree integral lattice and its middle determinant, but not the relations among e and v . Those relations will come from the first nonconstant critical manifold of the free-loop energy.

Write $\Lambda M = H^1(\mathbb{R}/T\mathbb{Z}, M)$ for the Sobolev free loop space, with rotation of the parameter as its circle action. The continuous-loop and Sobolev-loop models have the same homotopy type, compatibly with evaluation and rotation. We use the energy

$$\mathcal{E}(c) = \frac{1}{2} \int_0^T |\dot{c}(t)|^2 dt.$$

For a circle space A , write $A_{hS^1} = ES^1 \times_{S^1} A$. The constant-loop subspace has trivial action, hence its Borel construction is $M \times BS^1$.

Proposition 8.3 (The first attachment and its coefficient). *There is an oriented rank-three vector bundle $E \rightarrow B$ and an attaching map*

$$f : S(E) \longrightarrow M.$$

Let $p : S(E) \rightarrow B$ and put $x = f^*u \in H^4(S(E); \mathbb{Z})$. The following properties hold:

- (i) *The pullback ρ^*E is the first negative bundle for the free-loop energy. It is isomorphic as a real vector bundle to the rank-three bundle $K \rightarrow U$ with fiber*

$$K_w = \{a \in w^\perp : J_a(D) = 0\}, \quad J_a(0) = 0, \quad J'_a(0) = a,$$

where J_a is the normal Jacobi field along γ_w .

- (ii) *Let $\tilde{p} : S(\rho^*E) \rightarrow U$ and $\hat{p} : S(\rho^*E) \rightarrow S(E)$ be the pullback square. Then*

$$\hat{p}^*x = \tilde{p}^*z. \tag{8.10}$$

- (iii) *After choosing the orientation of E , one has*

$$p_!x = e. \tag{8.11}$$

With either orientation its coefficient is ± 1 .

- (iv) *The actual negative bundle satisfies*

$$\rho^*p_1(E) = 4z. \tag{8.12}$$

$$\begin{array}{ccccc}
S(\rho^*E) & \xrightarrow{\hat{\rho}} & S(E) & \xrightarrow{f} & M \\
\downarrow \tilde{p} \text{ } S^2 \text{ fibers} & & \downarrow p \text{ } S^2 \text{ fibers} & & \\
U = SM & \xrightarrow[\rho \text{ } (S^1 \text{ fibers})]{} & B = U/S^1 & & \\
\text{marked unit-speed} & & \text{oriented geodesic} & & \\
\text{oriented geodesic} & & \text{with no marked point} & &
\end{array}$$

$f \circ \hat{\rho} \simeq \pi \circ \tilde{p} \quad (\text{evaluation homotopy})$
 $x = f^*u, \quad z = \pi^*u, \quad \hat{\rho}^*x = \tilde{p}^*z$

Figure 2: The sphere bundle of the descended rank-three bundle E and its pullback to $U = SM$. The square is a pullback, with vertical fibers S^2 and horizontal fibers S^1 . The attaching map f is not point evaluation of an unmarked geodesic. Evaluation of the attachment gives the displayed homotopy and hence the equality of cohomology classes.

The attaching map makes $x^{r+1} = f^*(u^{r+1}) = 0$. The sphere integral $p_!x = e$ fixes the coefficient of the fiber part of x , while $\rho^*p_1(E) = 4z$ carries the incidence coefficient to the actual negative bundle. Separating x into base and fiber parts will turn its nilpotence into relations on B . The middle determinant will then fix the scale of the top-degree pairing used to evaluate N .

Figure 2 distinguishes the pullback square from the evaluation homotopy used in part (ii).

Proof. We give the index, attachment, and integral-coefficient details.

Critical manifolds and indices. The positive critical levels are $j^2T/2$, $j \geq 1$; the critical manifold at each is a copy of U , parametrized by $w \mapsto (t \mapsto \gamma_w(jt))$. These are all the nonconstant critical loops because every geodesic has prime length T .

For the j th iterate, reparametrize variations by arclength on $[0, jT]$. The kernel of the periodic Hessian consists of all periodic Jacobi fields. All normal Jacobi fields are T -periodic by the differential of $\phi_T = \text{Id}$, so their dimension is $2d$. A tangential Jacobi field is an affine scalar multiple of the velocity; the periodic boundary conditions force its scalar to be constant. The kernel therefore has dimension $2d + 1 = \dim U$. Variations of the initial unit tangent vector, including the time-shift direction, realize precisely these fields. Thus the Hessian kernel is the tangent space to the critical manifold, as required by the Morse–Bott condition.

The periodic index equals the fixed-endpoint index. To see this without an index correction formula, evaluation at the marked initial point maps the periodic kernel onto $T_{\pi(w)}M$: arbitrary normal initial values are possible, and the constant tangent field supplies the remaining direction. Choose a linear right inverse to this evaluation. Subtracting the corresponding kernel field from a periodic test field makes both endpoint values zero. The index form is unchanged, including its mixed terms, because a kernel field pairs to zero with every periodic test field by integration by parts. This subtraction is injective on every negative definite subspace. Conversely, fixed-endpoint test fields form a subspace of the periodic domain. The two indices are consequently equal.

Interior conjugate instants on $[0, jT]$ occur at the j odd multiples of D , with multiplicity three, and the $j - 1$ positive multiples of T , with multiplicity d . The fixed-endpoint Morse index theorem [15, Theorem 15.1, p. 83] gives

$$\lambda_j = 3j + d(j - 1). \quad (8.13)$$

In particular $\lambda_1 = 3$, while every later index is at least $d + 6 = n + 5 \geq 13$. The index count is unchanged on completing the piecewise smooth zero-endpoint fields in H^1 : the index form is

continuous, and any finite negative Gram matrix remains negative after sufficiently close smooth approximation.

Negative bundles and attachment. Let $\nu \rightarrow U$ be the negative spectral bundle of the periodic L^2 Jacobi operator

$$\mathcal{A}_w V = -\nabla_t^2 V - R(V, \dot{\gamma}_w) \dot{\gamma}_w.$$

This self-adjoint operator has domain the periodic H^2 fields, and its quadratic form is the index form on periodic H^1 fields. Its negative eigenvalues form an isolated finite cluster. The constant index and nullity, together with compactness of U , make the corresponding spectral projections locally smooth in the initial data. Elliptic regularity makes their eigenfields smooth, so ν is a genuine rank-three bundle. It represents the negative bundle for the Hilbert-space Morse–Bott attachment as well. Indeed, projection from the negative subspace of the bounded H^1 -Riesz Hessian to ν_w is injective: a vector in its kernel would have nonnegative index form in the L^2 spectral decomposition. Equal negative ranks make this a bundle isomorphism. Each fixed parameter rotation intertwines these operators and projections with those at the rotated geodesic. The resulting action on ν is continuous and covers the free smooth circle action on U . Hence ν descends to a real bundle $E \rightarrow B$, with $\nu \cong \rho^* E$. Since B is simply connected, E is orientable.

We use the Morse–Bott disk-attachment theorem for the energy on the H^1 loop space of a closed Riemannian manifold. The energy satisfies the Palais–Smale condition. Its Hessian is nondegenerate normal to each compact critical manifold just described, and has finite negative index. The bounded H^1 Hessian is Fredholm; its identified kernel therefore gives an invertible restriction normal to the critical manifold, as in the formulation cited below. Thus sublevels bracketing one critical value differ, up to homotopy, by the disk and sphere bundles of its negative bundle. This construction also respects parameter rotation. Indeed each fixed rotation is a smooth isometry for the natural loop-space metric, preserves energy, commutes with the gradient deformation, and preserves the Hessian splitting. The local normal form and the contraction of its positive directions may be constructed from these invariant metrics and splittings. Joint differentiability of the circle action on the whole H^1 manifold is not needed. For this equivariant negative-bundle formulation see Ottosen and Bökstedt [18, Section 7, pp. 2190–2192], following the Hilbert-space attachment of Klingenberg [12, Theorem 2.4.10]. The associated Borel vector bundle and Thom quotient are described in Ottosen and Bökstedt [18, Lemma 5.1, pp. 2185–2186].

Here is the attaching map and its evaluation property explicitly. Start a sufficiently small negative disk at every first-level geodesic by pointwise exponential of the negative fields. Its boundary lies strictly below the first positive critical value, uniformly over U . That lower sublevel deforms equivariantly to the constants. This follows from the noncritical energy deformation between the boundary level and a small positive level, followed by contraction of small loops to constants. For the latter, embed M in Euclidean space, average a sufficiently short loop, and project the straight contraction to a tubular neighborhood of M . Small energy gives uniformly small length, so this is defined and continuous, preserves the constants, and commutes with rotation. The projection has bounded derivative, so choosing the initial small level sufficiently small keeps this contraction below the first critical value.

Insert the boundary deformation in an outer collar of the small negative disk bundle. This produces an equivariant map of pairs

$$\mathcal{F}_1 : (D(\nu), S(\nu)) \longrightarrow (\Lambda M, M)$$

whose zero section is the first critical manifold and which represents its Morse–Bott attachment. On the boundary it takes values in the constants. Because the action on the constants is trivial, this boundary map descends to $f : S(E) \rightarrow M$. Evaluation at time zero composed with $\mathcal{F}_1$ is homotopic, by radial contraction of the disk fibers, to π composed with their bundle projection. Restrict this homotopy to $S(\nu) \cong S(\rho^* E)$. Its boundary value is $f \circ \hat{\rho}$, which proves (8.10).

Choose regular sublevels Λ_- and Λ_+ immediately below and above the first positive critical value. The preceding attachment gives, over $\mathbb{Z}$ or any field $\mathbb{F}$,

$$H^q(\Lambda_+, M; \mathbb{F}) \cong H^{q-3}(U; \mathbb{F}), \quad (8.14)$$

$$H^q((\Lambda_+)_{hS^1}, M \times BS^1; \mathbb{F}) \cong H^{q-3}(B; \mathbb{F}). \quad (8.15)$$

These are Thom isomorphisms; in the second one the free first critical action identifies the Borel negative bundle with E up to homotopy. One can take finite-dimensional free approximations to ES^1 throughout, apply the invariant attachment, and then pass to the limit. For every later critical level, the Borel negative bundle still exists over the Borel critical manifold even though the action has finite stabilizers. Its relative cohomology vanishes below its rank λ_j , with any necessary orientation local system. The same statement holds without the Borel construction. Equation (8.13) therefore shows that subsequent attachments do not change cohomology in degrees at most five. The exhaustion by sublevels causes no extra term in these degrees: the restriction systems in the relevant degrees are eventually constant, so their inverse-limit obstruction vanishes.

We have constructed the attachment and isolated the degrees in which no later attachment can alter it. What remains is to fix its coefficient integrally and identify its actual negative bundle.

The coefficient is a unit over every prime field. Since $H^2(B; \mathbb{Z}) = \mathbb{Z}e$, write $p_!x = ce$ with $c \in \mathbb{Z}$. Under (8.15), the connecting homomorphism

$$H^4(M \times BS^1; \mathbb{F}) \longrightarrow H^5((\Lambda_+)_{hS^1}, M \times BS^1; \mathbb{F}) \cong H^2(B; \mathbb{F})$$

sends the class u from M to $p_!x$, up to the Thom orientation sign. Indeed the connecting map first restricts to the attaching sphere, where that class is x , and then uses the boundary map for $(D(E), S(E))$; the inverse Thom isomorphism identifies this last map with integration over the oriented sphere fiber.

Suppose that c vanished modulo a prime ℓ , and put $\mathbb{F} = \mathbb{Z}/\ell$. Exactness would extend the pure class $u \in H^4(M \times BS^1; \mathbb{F})$ over $(\Lambda_+)_{hS^1}$ and hence over $(\Lambda M)_{hS^1}$, by the higher-index gap. After forgetting equivariance, its restriction would be the evaluation class

$$a = \text{ev}_0^* u \in H^4(\Lambda M; \mathbb{F}).$$

To justify uniqueness here, (8.14) gives $H^4(\Lambda_+, M; \mathbb{F}) = H^1(U; \mathbb{F}) = 0$. Higher attachments have no effect in this range, so restriction $H^4(\Lambda M; \mathbb{F}) \rightarrow H^4(M; \mathbb{F})$ is injective. The evaluation class is already one extension of u , and thus is the only one.

An ordinary class restricted from the Borel construction has zero action slant. For completeness, if Y is any circle space, fix $e_0 \in ES^1$ and let $i(y) = [e_0, y] \in Y_{hS^1}$. For the action map $A : S^1 \times Y \rightarrow Y$, the two maps $i \circ A$ and $i \circ \text{pr}_Y$ are homotopic:

$$[e_0, gy] = [e_0g, y],$$

and the loop $g \mapsto e_0g$ contracts in ES^1 . Consequently the pullback of a Borel class under $i \circ A$ has no component in $H^1(S^1; \mathbb{F}) \otimes H^{*-1}(Y; \mathbb{F})$. Equivalently its slant with $[S^1]$ is zero. This is also the usual action description of the Borel d_2 differential.

The evaluation class a has a nonzero action slant over every prime field. The integral cohomology ring of M and the universal coefficient theorem give $H_2(M; \mathbb{Z}) = H_3(M; \mathbb{Z}) = 0$. Simple connectivity and successive applications of the Hurewicz theorem [9, Theorem 4.32] make M three-connected; Hurewicz then identifies $\pi_4(M)$ with $H_4(M; \mathbb{Z}) = \mathbb{Z}$. Choose a based map $g : S^4 \rightarrow M$ with $\langle g^*u, [S^4] \rangle = 1$, and let $\hat{g} : S^3 \rightarrow \Omega M \subset \Lambda M$ be its adjoint. Evaluation of the

rotated based loops on $S^1 \times S^3$ factors, up to the orientation convention, through the smash quotient

$$S^1 \times S^3 \longrightarrow S^1 \wedge S^3 = S^4 \xrightarrow{g} M.$$

The quotient takes the cross-product fundamental class to the fundamental class of S^4 , up to sign. Therefore

$$\langle \widehat{g}^*(A^* \text{ev}_0^* u / [S^1]), [S^3] \rangle = \pm 1$$

integrally, and remains nonzero after reduction modulo ℓ . This contradicts the zero slant required of the hypothesized Borel extension. No prime divides c , so $c = \pm 1$. Reverse the orientation of E if necessary to obtain (8.11).

The negative bundle is the cut-fiber tangent bundle. For $a \in K_w$, truncate J_a to the first half-period and extend it by zero on the second half:

$$\widehat{J}_a(t) = \begin{cases} J_a(t), & 0 \leq t \leq D, \\ 0, & D \leq t \leq T. \end{cases}$$

The endpoint values agree, so this is an H^1 periodic test field. Integration by parts on $[0, D]$, with the vanishing endpoint values and the Jacobi equation, gives zero index form on this entire three-dimensional space. Project these fields to the negative spectral space ν_w . If a projected vector were zero, its zero index form and its membership in the nonnegative spectral subspace would force it into the Hessian kernel. It would then be a smooth periodic Jacobi field. But a nonzero J_a has nonzero derivative at D , since both zero value and zero derivative would make it identically zero. Its truncation has a derivative jump at D and cannot be a kernel field. The projection is thus injective, and equal ranks make it an isomorphism $K_w \rightarrow \nu_w$. The truncation depends continuously in the H^1 norm on w and a , and the spectral projections vary continuously as above. We obtain an isomorphism of real bundles $K \cong \nu \cong \rho^* E$. The truncations are used in the H^1 form domain, not in the H^2 operator domain; their derivative jump is the reason that a nonzero truncation cannot lie in the smooth kernel.

Finally define the full cut map

$$\mathfrak{a} : U \longrightarrow F, \quad \mathfrak{a}(w) = (\pi(w), \exp_{\pi(w)}(Dw)).$$

By the actual bundle identification in Proposition 6.1, its pullback of $\mathcal{V}$ is the initial-direction four-plane. The marked unit direction w gives a trivial line in this plane; its orthogonal complement is exactly K_w , the tangent space of the great-three-sphere fiber. Hence

$$\mathfrak{a}^* \mathcal{V} \cong \mathbb{R} \oplus K \cong \mathbb{R} \oplus \rho^* E.$$

The two point maps from U to M in $\mathfrak{a}$ are π and $\pi \circ \phi_D$. The geodesic flow provides a homotopy between them, so $\mathfrak{a}^* u_1 = \mathfrak{a}^* u_2 = z$. Taking first Pontryagin classes and using (8.1) proves (8.12). $\square$

8.3 Half-integral lifts and the middle pairing

On each sphere fiber, half the Euler class of its tangent bundle integrates to one. This gives a normalized fiber class with which to separate x into base and fiber parts. The base class may be only half-integral; we will track its displacement from the integral lift v so that the middle determinant is preserved.

Proposition 8.4 (Arithmetic parameters). *Let $p : S(E) \rightarrow B$ and x be as in Proposition 8.3, and let*

$$\xi = \frac{1}{2} e(T_{\text{vert}} S(E)) \in H^2(S(E); \mathbb{Q}), \quad X = e^2 \in H^4(B; \mathbb{Z}).$$

There is a class $Y \in H^4(B; \mathbb{Q})$ such that, with pullbacks by p suppressed,

$$x = Y + e\xi, \quad \xi^2 = p_1(E)/4, \quad p_!\xi = 1, \quad \rho^*Y = z, \quad 2Y \in H^4(B; \mathbb{Z}). \quad (8.16)$$

For an integer b ,

$$Y = v + \frac{b}{2}e^2. \quad (8.17)$$

There are rational numbers α, β with

$$L := e^2 p_1(E)/4 = \alpha X^2 + \beta XY, \quad 4\alpha \in \mathbb{Z}, \quad |\beta| = 1. \quad (8.18)$$

Here $\beta = 1$ for (8.1); retaining β also allows the opposite generator convention, for which $\beta = -1$. Moreover

$$(Y + e\xi)^{r+1} = (Y - e\xi)^{r+1} = 0 \quad \text{in } H^*(S(E); \mathbb{Q}). \quad (8.19)$$

Count X, Y as polynomial degree one, corresponding to cohomological degree four. For homogeneous polynomials of degree $2r - 1$, set

$$\ell(P) = \int_B eP(X, Y).$$

Then the middle moment matrix satisfies

$$\left| \det \left[\ell(X^{2r-1-i-j} Y^{i+j}) \right]_{0 \leq i, j < r} \right| = r + 1, \quad \ell(X^{2r-1}) = N. \quad (8.20)$$

Pullback $p^* : H^*(B; \mathbb{Q}) \rightarrow H^*(S(E); \mathbb{Q})$ is injective, and multiplication by e on $H^n(B; \mathbb{Q})$ is injective.

Proof. Give each unit sphere fiber the boundary orientation from E . The marked unit vector in p^*E spans a trivial line, with orthogonal complement the oriented vertical tangent two-plane:

$$p^*E \cong \mathbb{R} \oplus T_{\text{vert}}S(E).$$

For an oriented real two-plane bundle, the first Pontryagin class is the square of its Euler class. Thus $\xi^2 = p^*p_1(E)/4$. On a sphere fiber the Euler class of its tangent bundle integrates to two, giving $p_!\xi = 1$.

The Euler class of the oriented rank-three bundle E lies in $H^3(B; \mathbb{Z}) = 0$, so it vanishes. The sphere-bundle Gysin sequence then gives, in every degree, a short exact sequence whose first map is p^* and whose last map is $p_!$. In particular p^* is injective. Over $\mathbb{Q}$, the classes $1, \xi$ restrict to a basis of each sphere's cohomology, so Leray–Hirsch gives the module decomposition

$$H^*(S(E); \mathbb{Q}) = p^*H^*(B; \mathbb{Q}) \oplus \xi p^*H^{*-2}(B; \mathbb{Q}).$$

Since $p_!x = e$, there is a unique rational class Y with $x = p^*Y + p^*e\xi$. The integral class

$$2x - p^*e e(T_{\text{vert}}S(E))$$

has zero sphere integral. Integral Gysin exactness expresses it as p^*Y_2 for $Y_2 \in H^4(B; \mathbb{Z})$. Rational injectivity of p^* gives $Y_2 = 2Y$. This proves half-integrality without assuming that Y itself is integral.

Pull back to the square over U in Proposition 8.3. The class ρ^*e vanishes, as it does for the Euler class pulled to its own circle bundle. Equation (8.10) now gives

$$\tilde{p}^*(\rho^*Y) = \tilde{\rho}^*x = \tilde{p}^*z.$$

The pullback sphere bundle also has zero Euler class; its rational Gysin sequence makes $\widehat{p}^*$ injective. It follows that $\rho^*Y = z$. Write $2Y = av + be^2$ in the integral basis supplied by Lemma 8.2. Since $\rho^*v = z$ is a primitive generator of $H^4(U; \mathbb{Z})$, pulling back gives $a = 2$. This proves (8.17).

Similarly, integrality of $p_1(E)$ and (8.12) give

$$p_1(E) = Ae^2 + 4v, \quad A \in \mathbb{Z}.$$

If the opposite sign of the incidence coefficient is retained, this reads $p_1(E) = Ae^2 + 4\beta v$, with $\beta = \pm 1$. In either convention substitution of $v = Y - (b/2)e^2$ gives

$$L = \frac{A - 2\beta b}{4}X^2 + \beta XY.$$

Thus $4\alpha = A - 2\beta b$ is integral, proving (8.18).

The first equality in (8.19) is $x^{r+1} = f^*(u^{r+1}) = 0$. Fiber antipodal reflection covers the identity of B and reverses the orientation of the two-dimensional sphere fibers. Its pullback fixes base classes and sends the vertical Euler class, hence ξ , to its negative. Pulling back the first equality proves the second one.

It remains to identify the moment determinant. An integral basis of $H^{n-2}(B; \mathbb{Z})$ from (8.4) is

$$b_i = eX^{r-1-i}v^i, \quad 0 \leq i < r.$$

The matrix of (8.7) in this basis has entries

$$\int_B eb_i b_j = \int_B eX^{2r-1-i-j}v^{i+j}.$$

Replace v in these basis vectors by $Y = v + (b/2)X$. The change of basis over $\mathbb{Q}$ is triangular with every diagonal entry equal to one. Hence its determinant is one and it does not change the determinant of the bilinear form. Lemma 8.2 proves the first assertion in (8.20). The second follows from $eX^{2r-1} = e^{4r-1}$ and the definition of N . The final injectivity statements were proved above for p^* and in Lemma 8.2 for multiplication by e . $\square$

9 The sharp quaternionic volume bound

We retain the notation of Section 8. Thus $n = 4r$, $r \geq 2$, and

$$X = e^2, \quad L = e^2 p_1(E)/4 = \alpha X^2 + \beta XY, \quad 4\alpha \in \mathbb{Z}, \quad |\beta| = 1.$$

In this section the polynomial degrees of X and Y are both one; one polynomial degree therefore means cohomological degree four. Let

$$\ell(P) = \int_B eP(X, Y), \quad \deg P = 2r - 1. \quad (9.1)$$

The inputs of Proposition 8.4 include

$$I_j = \ell(X^{2r-1-j}Y^j), \quad |\det(I_{i+j})_{0 \leq i, j < r}| = r + 1, \quad N = I_0 > 0. \quad (9.2)$$

Our goal is $N \leq \binom{4r}{2r}/(2r + 1)$, the model value computed in Proposition 8.1. The attachment gives polynomial relations; the middle determinant fixes the scale of the top functional; an explicit coefficient sum then gives the bound. We do not prescribe the cohomology ring of the geodesic quotient in advance.

9.1 The power relations and the nonzero parameter

Introduce a formal variable Z with $Z^2 = L$ and define homogeneous polynomials in X, Y by

$$Q(X, Y) = \frac{(Y + Z)^{r+1} - (Y - Z)^{r+1}}{2Z} = \sum_{j \geq 0} \binom{r+1}{2j+1} Y^{r-2j} L^j, \quad (9.3)$$

$$R(X, Y) = \frac{(Y + Z)^{r+1} + (Y - Z)^{r+1}}{2} = \sum_{j \geq 0} \binom{r+1}{2j} Y^{r+1-2j} L^j. \quad (9.4)$$

Only the terms with valid binomial indices are included. The quotient in (9.3) is a polynomial identity; it involves no division by a cohomology class. The degrees of Q, R are $r, r+1$, respectively.

Lemma 9.1. *The classes $Q(X, Y)$ and $R(X, Y)$ vanish in $H^*(B; \mathbb{Q})$. Moreover $\alpha \neq 0$, and consequently $|4\alpha| \geq 1$.*

Proof. On the sphere bundle $p : S(E) \rightarrow B$, the attachment class is $x = Y + e\xi$ and satisfies $x^{r+1} = 0$. The fiber antipode sends ξ to $-\xi$, so $(Y - e\xi)^{r+1} = 0$ as well. Taking the sum gives $p^*R = 0$. The rational Leray–Hirsch decomposition with basis $1, \xi$ makes p^* injective, hence $R = 0$. Taking the difference and integrating over the sphere fiber, using $p_!\xi = 1$, gives $eQ = 0$. The class Q has cohomological degree $4r = n$, and multiplication by e is injective from $H^n(B; \mathbb{Q})$ to $H^{n+2}(B; \mathbb{Q})$ by Lemma 8.2. Therefore $Q = 0$.

Suppose now that $\alpha = 0$. We use the nondegenerate moment pairing in (9.2) to obtain a contradiction. Write

$$q(y) = Q(1, y), \quad a(y) = R(1, y), \quad A_0 = \mathbb{C}[y]/(q).$$

The polynomial q has degree r and leading coefficient $r+1$. The relations $Q = 0$ imply that the rule $y^j \mapsto I_j$, for $0 \leq j \leq 2r-1$, agrees with reduction modulo q followed by a functional $F : A_0 \rightarrow \mathbb{C}$. Indeed, every multiple of q needed in dividing a polynomial of this degree has multiplier of degree at most $r-1$, and its homogenization is a top-degree multiple of Q . Thus the matrix of $(f, g) \mapsto F(fg)$ in $1, y, \dots, y^{r-1}$ is precisely the nonsingular matrix in (9.2). The relation $R = 0$ similarly gives

$$F(a(y)y^j) = 0 \quad (0 \leq j \leq r-2). \quad (9.5)$$

Let $\varepsilon : A_0 \rightarrow \mathbb{C}$ extract the coefficient of y^{r-1} in the remainder of degree less than r . Its product pairing is nondegenerate: the entry $\varepsilon(y^{i+j})$ is zero when $i+j < r-1$ and one when $i+j = r-1$, for $0 \leq i, j < r$. Reversing the columns gives a triangular matrix with diagonal entries one. Hence there is a unique $b \in A_0$ such that $F(f) = \varepsilon(bf)$ for all f . Nondegeneracy of the F pairing means that multiplication by b is invertible, so b is a unit. For the ε pairing the orthogonal complement of $1, y, \dots, y^{r-2}$ consists exactly of the constants. To see this, write a remainder $c_0 + \dots + c_{r-1}y^{r-1}$ and successively test against $1, y, \dots, y^{r-2}$; the tests force $c_{r-1}, c_{r-2}, \dots, c_1$ to vanish in that order. Equation (9.5) therefore says that ba is constant. In particular any nonzero class a satisfying that equation must be a unit: its product with the unit b is a nonzero constant.

But the class of $a = R(1, y)$ is nonzero and is not a unit. Both q and a vanish at $y = 0$ when $\alpha = 0$, so evaluation at zero precludes invertibility. To check that the class is nonzero, choose an $(r+1)$ st root of unity $\eta \neq 1, -1$, and put

$$c_\eta = \frac{\eta - 1}{\eta + 1}, \quad y_0 = \frac{\beta}{c_\eta^2}, \quad Z_0 = c_\eta y_0.$$

These numbers are nonzero, $Z_0^2 = \beta y_0$, and $y_0 + Z_0 = \eta(y_0 - Z_0)$ with $y_0 - Z_0 \neq 0$. The formulas for Q, R give $q(y_0) = 0$ but $a(y_0) = (y_0 - Z_0)^{r+1} \neq 0$. Thus a cannot be zero modulo q , a contradiction. We conclude that $\alpha \neq 0$. Its stated integrality then gives $|4\alpha| \geq 1$. $\square$

Lemma 9.2. *For $\alpha \neq 0$, the polynomials Q, R have no common nonzero zero over $\mathbb{C}$. The algebra*

$$\mathcal{A} = \mathbb{C}[X, Y]/(Q, R)$$

has Hilbert series

$$\frac{(1-t^r)(1-t^{r+1})}{(1-t)^2} = (1+t+\cdots+t^{r-1})(1+t+\cdots+t^r). \quad (9.6)$$

In particular its top degree is $2r-1$, with dimension one, and

$$X : \mathcal{A}_{r-1} \longrightarrow \mathcal{A}_r \quad \text{is an isomorphism,} \quad \dim \mathcal{A}_{r-1} = \dim \mathcal{A}_r = r. \quad (9.7)$$

Furthermore the complete rational relation ideal of the classes e, Y is generated by $Q(e^2, Y)$ and $R(e^2, Y)$.

Proof. At a common zero (X, Y) choose a complex square root Z of $\alpha X^2 + \beta XY$. If $Z \neq 0$, simultaneous vanishing of Q, R would force both $(Y+Z)^{r+1}$ and $(Y-Z)^{r+1}$ to vanish, which is impossible unless $Y = Z = 0$. If $Z = 0$, the formulas become $Q = (r+1)Y^r$ and $R = Y^{r+1}$, so $Y = 0$; then $\alpha X^2 = 0$ forces $X = 0$ as well.

The two homogeneous polynomials are therefore relatively prime. Multiplication by Q in $\mathbb{C}[X, Y]$ is injective, and multiplication by R is injective modulo Q : in the polynomial unique factorization domain, $Q \mid Rf$ and relative primality imply $Q \mid f$. The two resulting multiplication exact sequences give (9.6). On setting $X = 0$, the quotient becomes $\mathbb{C}[Y]/(Y^r)$. Thus the degree- r cokernel of multiplication by X vanishes. The two dimensions read from (9.6) are equal, which proves (9.7).

For the assertion about the whole cohomology ring, Lemma 8.2 gives generation of $H^*(B; \mathbb{Q})$ by e, v and total dimension $2r(r+1)$. Since $Y = v + (b/2)e^2$ by (8.17), the same ring is generated by e, Y .

The polynomial algebra $\mathbb{C}[e, Y]$ is free on $1, e$ over $\mathbb{C}[X, Y]$ under $X = e^2$. Quotienting by the two displayed relations therefore gives $\mathcal{A} \oplus e\mathcal{A}$, of dimension $2r(r+1)$ by (9.6). Its surjection to $H^*(B; \mathbb{C})$ is an isomorphism by the equality of dimensions. The relations and maps are defined over $\mathbb{Q}$, giving the rational assertion as well. $\square$

9.2 An explicit top functional and its determinant

The algebraic method is the classical perfect-pairing duality for finite graded complete intersections; see Huneke [10, Corollary 3.5 and the discussion following Theorem 9.1]. We make the pairing explicit to determine its normalization and the required determinant.

The power relations suggest replacing $Y+Z$ and $Y-Z$ by two nilpotent variables a_1, a_2 . With $Y = (a_1 + a_2)/2$ and $Z = (a_1 - a_2)/2$, the equation $Z^2 = L$ becomes a quadratic relation for X . The resulting algebra admits explicit coefficient extraction; the involution exchanging a_1, a_2 will recover the pairing on $\mathcal{A}$ without cancelling Z .

For any complex $\alpha \neq 0$ and any complex β , set

$$\mathcal{D}_{\alpha, \beta} = \frac{\mathbb{C}[a_1, a_2, X]}{(a_1^{r+1}, a_2^{r+1}, 4\alpha X^2 + 2\beta(a_1 + a_2)X - (a_1 - a_2)^2)}. \quad (9.8)$$

All three variables have polynomial degree one. Every class has a unique remainder that has X -degree at most one and each a_i -degree at most r . Denote extraction of $X a_1^r a_2^r$ in that remainder by $\varepsilon_{\mathcal{D}}$. On homogeneous polynomials of degree $2r-1$ define

$$\Phi_{\alpha, \beta}(P) = \varepsilon_{\mathcal{D}} \left((a_1 - a_2)^2 P \left(X, \frac{a_1 + a_2}{2} \right) \right). \quad (9.9)$$

Lemma 9.3. *The functional $\Phi_{\alpha,\beta}$ descends to the top degree of $\mathcal{A}$ and is nonzero. If*

$$J_j = \Phi_{\alpha,\beta}(X^{2r-1-j}Y^j),$$

then for every $\alpha \neq 0$ and every $\beta \in \mathbb{C}$,

$$\det(J_{i+j})_{0 \leq i,j < r} = (-1)^{r(r+1)/2} (r+1)(4\alpha)^{-r(r-1)/2}. \quad (9.10)$$

For the geometric parameters there is consequently a scalar K with

$$\ell = K\Phi_{\alpha,\beta}, \quad |K| = |4\alpha|^{(r-1)/2}. \quad (9.11)$$

Proof. Put $Y = (a_1 + a_2)/2$ and $Z = (a_1 - a_2)/2$ inside $\mathcal{D}$. Its quadratic relation becomes $Z^2 = L$ and its other relations become

$$R = 0, \quad ZQ = 0.$$

It follows at once that multiplying either R or Q by $4Z^2$ gives zero, so (9.9) annihilates the top-degree multiples of both relations.

We next prove nondegeneracy, uniformly in the complex parameter β . The coefficient functional on

$$D_0 = \mathbb{C}[a_1, a_2]/(a_1^{r+1}, a_2^{r+1})$$

extracting $a_1^r a_2^r$ has a perfect product pairing, since every monomial pairs with its complementary monomial. The quadratic relation in (9.8) is monic after division by 4α , so $\mathcal{D} = D_0 \oplus XD_0$. Its coefficient pairing is perfect as well. Indeed, if $f + Xg$ annihilates every element of D_0 , extraction shows that $g = 0$; testing then against XD_0 shows that $f = 0$. The grading makes this a perfect pairing between complementary degrees summing to $2r + 1$.

The involution exchanging a_1, a_2 preserves $\varepsilon_{\mathcal{D}}$. Its even and odd subspaces are therefore orthogonal, and the odd subspaces in complementary degrees pair perfectly. Their exact module description is particularly useful. Before imposing R, ZQ , the relation $Z^2 = L$ gives the direct sum

$$\mathbb{C}[X, Y] \oplus Z\mathbb{C}[X, Y].$$

The ideal generated by R, ZQ has even part (R, LQ) and odd part $Z(R, Q)$. Consequently

$$\mathcal{D}^- \simeq Z\mathcal{A} \quad \text{as a graded module, with a shift by one.} \quad (9.12)$$

This follows by taking the two summands of the ideal; no cancellation of Z in the quotient is used.

By (9.7), multiplication by X carries $\mathcal{D}_r^-$ isomorphically onto $\mathcal{D}_{r+1}^-$. In the basis

$$(a_1 - a_2)X^{r-1-i}Y^i, \quad 0 \leq i < r,$$

of $\mathcal{D}_r^-$, the bilinear pairing $(f, g) \mapsto \varepsilon_{\mathcal{D}}(fXg)$ has matrix (J_{i+j}) . It is nondegenerate because the odd complementary-degree pairing is perfect. This proves in particular that Φ is nonzero.

For fixed nonzero α , reduction of powers of X by its monic quadratic relation makes every J_j a polynomial in β . The determinant is thus a polynomial in β which, by the preceding argument, has no complex zero. It is a nonzero constant in β . It remains to evaluate it at $\beta = 0$.

First set $\alpha = 1/4$ as well, so that $X^2 = (a_1 - a_2)^2$. In $\mathcal{A}$ use the linear coordinates

$$A_1 = Y + X/2, \quad A_2 = Y - X/2.$$

For $p + q = 2r - 1$, the coefficient of X in the quadratic remainder of $A_1^p A_2^q$ is the polynomial

$$\frac{a_1^p a_2^q - a_1^q a_2^p}{2(a_1 - a_2)}.$$

The numerator is divisible by $a_1 - a_2$. Multiplying by $(a_1 - a_2)^2$ and extracting $a_1^r a_2^r$ gives

$$\Phi_{1/4,0}(A_1^p A_2^q) = \begin{cases} -1, & (p, q) = (r, r-1), \\ 1, & (p, q) = (r-1, r), \\ 0, & \text{otherwise.} \end{cases} \quad (9.13)$$

In the degree- $(r-1)$ monomial basis $A_1^{r-1-i} A_2^i$, the matrix of the middle $X = A_1 - A_2$ pairing has entries -2 for $i+j = r-1$, entries 1 for $i+j = r-2$ or r , and zero otherwise. Reversing columns gives the tridiagonal matrix with diagonal -2 and adjacent entries 1 . Its size- r determinant is $(-1)^r(r+1)$: expansion along the last row gives the recurrence $d_r = -2d_{r-1} - d_{r-2}$ with $d_0 = 1, d_1 = -2$. Column reversal has sign $(-1)^{r(r-1)/2}$. The linear change from X, Y to A_1, A_2 has determinant one, so its degree- $(r-1)$ monomial change also has determinant one. We have proved (9.10) at $(\alpha, \beta) = (1/4, 0)$.

At $\beta = 0$ for general $\alpha \neq 0$, odd-indexed J_j vanish. For $j = 2h$, quadratic reduction gives

$$J_{2h}(\alpha, 0) = (4\alpha)^{-(r-1-h)} J_{2h}(1/4, 0).$$

Every nonzero permutation term in the size- r determinant therefore has parameter exponent

$$-\sum_{i=0}^{r-1} \left(r-1 - \frac{i + \sigma(i)}{2} \right) = -\frac{r(r-1)}{2}.$$

This proves (9.10) for $\beta = 0$, and its independence of β proves the full identity.

Finally, ℓ and Φ are nonzero functionals on the same one-dimensional top degree of $\mathcal{A}$, so $\ell = K\Phi$. Taking absolute determinants in (9.2) and (9.10) gives

$$r+1 = |K|^r (r+1) |4\alpha|^{-r(r-1)/2},$$

which is (9.11). □

9.3 The model upper bound

Theorem 9.4. *For a quaternionic-type Blaschke manifold of dimension $4r$, $r \geq 2$, the contact characteristic number of Proposition 8.1 satisfies*

$$N = \int_B e^{4r-1} \leq \frac{1}{2r+1} \binom{4r}{2r}. \quad (9.14)$$

Proof. The quadratic relation in (9.8) is

$$X^2 = \mathbf{a}X + \mathbf{b}, \quad \mathbf{a} = -\frac{\beta}{2\alpha}(a_1 + a_2), \quad \mathbf{b} = \frac{(a_1 - a_2)^2}{4\alpha}.$$

The coefficient of X in the remainder of X^q satisfies $D_1 = 1$, $D_2 = \mathbf{a}$, and $D_q = \mathbf{a}D_{q-1} + \mathbf{b}D_{q-2}$. Induction on this recurrence yields

$$D_{2r-1} = \sum_{j=0}^{r-1} \binom{2r-2-j}{j} \mathbf{a}^{2r-2-2j} \mathbf{b}^j.$$

Consequently, with ordinary polynomial coefficient extraction,

$$\Phi_{\alpha,\beta}(X^{2r-1}) = \sum_{j=0}^{r-1} \binom{2r-2-j}{j} \left(-\frac{\beta}{2\alpha} \right)^{2r-2-2j} (4\alpha)^{-j} c_j, \quad (9.15)$$

where

$$c_j = [a_1^r a_2^r](a_1 + a_2)^{2r-2-2j}(a_1 - a_2)^{2j+2}.$$

The exact sign of this central coefficient is

$$\text{sign}(c_j) = (-1)^{j+1}. \quad (9.16)$$

Indeed, substitute $a_1 = e^{it}$, $a_2 = e^{-it}$ and average over a full period. The coefficient becomes $(-1)^{j+1}$ times the strictly positive integral of a positive constant times $\cos^{2r-2-2j}(t) \sin^{2j+2}(t)$.

By Lemmas 9.1 and 9.3, the parameter factor in the absolute value of each summand of $K\Phi_{\alpha,\beta}(X^{2r-1})$, apart from its binomial and central coefficients, is

$$\begin{aligned} |K| \left| \frac{\beta}{2\alpha} \right|^{2r-2-2j} |4\alpha|^{-j} &= 2^{2r-2-2j} |4\alpha|^{j-3(r-1)/2} \\ &\leq 2^{2r-2-2j}. \end{aligned}$$

Here $|\beta| = 1$, $|4\alpha| \geq 1$, and $j - 3(r-1)/2 \leq -(r-1)/2 < 0$. The bound on the right is attained by the formal parameters $\alpha = -1/4$, $\beta = 1$, for which the normalization factor $|4\alpha|^{(r-1)/2}$ is one. At those parameters (9.16) shows that all the summands in (9.15) have the same, negative, sign. The sum of the termwise absolute bounds therefore equals the absolute value of that single formal evaluation.

We evaluate it explicitly. When $\alpha = -1/4$, $\beta = 1$, the two formal roots of the quadratic in X are $(\sqrt{a_1} + \sqrt{a_2})^2$ and $(\sqrt{a_1} - \sqrt{a_2})^2$. Their divided difference gives the polynomial

$$\begin{aligned} D_{2r-1} &= \frac{(\sqrt{a_1} + \sqrt{a_2})^{4r-2} - (\sqrt{a_1} - \sqrt{a_2})^{4r-2}}{4\sqrt{a_1 a_2}} \\ &= \frac{1}{2} \sum_{h=0}^{2r-2} \binom{4r-2}{2h+1} a_1^{2r-2-h} a_2^h. \end{aligned}$$

Thus multiplication by $(a_1 - a_2)^2$ and central coefficient extraction give

$$\left| \Phi_{-1/4,1}(X^{2r-1}) \right| = \binom{4r-2}{2r-1} - \binom{4r-2}{2r-3} = \frac{1}{2r+1} \binom{4r}{2r}.$$

Since $N = \ell(X^{2r-1}) > 0$, the preceding termwise comparison proves (9.14). $\square$

10 Completion of the metric classification

Theorem 1.2 reduces metric rigidity to a volume upper bound at the same diameter. The intrinsic quaternionic bound has now been proved. We first check the precise classical inputs for the complex and plane cases, then assemble the model list. For $\mathbb{HP}^2$, the classical plane argument is an alternative to the intrinsic bound, which already includes every quaternionic rank $r \geq 2$.

Corollary 10.1 (Complex-projective type). *A Blaschke manifold of complex-projective type is isometric to its equally scaled standard complex projective model.*

Proof. Write its real dimension as $2r$, with $r \geq 2$; the projective line is a sphere. Its integral cohomology ring is $\mathbb{Z}[u]/(u^{r+1})$, where u has degree two, and it is simply connected. Represent u by a map to $\mathbb{CP}^\infty$ [9, Theorem 4.57]. A closed smooth $2r$ -manifold has the homotopy type of a finite $2r$ -dimensional CW complex, so cellular approximation [9, Theorem 4.8] places the map in the $2r$ -skeleton $\mathbb{CP}^r$. Pullback identifies the integral cohomology generators and all their powers. The universal coefficient theorem for these finite CW complexes then gives an integral homology isomorphism. The homology Whitehead theorem for simply connected CW complexes [9, Corollary 4.33] makes this map a homotopy equivalence. Yang's volume theorem [28, Theorem, p. 380] therefore gives equality with the model at the same prime geodesic period, equivalently the same diameter. Corollary 1.3 applies. $\square$

Corollary 10.2 (Quaternionic and Cayley planes). *A Blaschke manifold with model $\mathbb{HP}^2$ or $\mathbb{OP}^2$ is isometric to that model at the same diameter.*

Proof. The projective-plane structure of such a Blaschke manifold is given by McKay [14, Lemma 5]. The classification of smooth topological projective planes [13, Corollary C] makes M diffeomorphic to its standard model. Both metrics are Zoll: all their unit-speed geodesics are simply closed with the same prime length $T = 2D$. After a common rescaling to $T = \pi$, the Riemannian weak Blaschke volume theorem recorded by Wilking and attributed there to Reznikov applies: a manifold with this common least period that is homeomorphic to a compact rank-one symmetric space has the same volume as its model [27, p. 282]. The diffeomorphism above supplies its homeomorphism hypothesis. Thus $V = V_0$, and this equality persists when the common rescaling is undone. Corollary 1.3 applies. $\square$

Corollary 10.3 (Quaternionic type). *A Blaschke manifold of quaternionic type is isometric to its standard quaternionic model at the same diameter.*

Proof. For $\mathbb{HP}^r$ with $r \geq 2$, use Theorem 9.4 and Proposition 8.1. At the common period $T = \pi$, the contact formula has the same positive volume factor for M and its model, so

$$\frac{N}{N_0} = \frac{\text{Vol}(M, g)}{\text{Vol}(\mathbb{HP}^r)}.$$

Here N_0 is the model number in (8.3), and the model volume computed there is

$$\text{Vol}(\mathbb{HP}^r) = \frac{\pi^{2r}}{(2r+1)!}.$$

Theorem 9.4 gives $N \leq N_0$, hence $\text{Vol}(M, g) \leq \text{Vol}(\mathbb{HP}^r)$. Corollary 1.3 proves the metric statement. The projective line is already the sphere case. $\square$

Proof of Theorem 1.1. Lemma 2.1 gives the common minimizing distance from the stated global injectivity-radius equality. The one-dimensional, sphere-type and real-projective-type cases are covered by Section 2. The classical structure list leaves complex type, quaternionic type, and the Cayley plane. These are respectively Corollaries 10.1, 10.3, and 10.2. Each conclusion is a metric isometry with the model at equal diameter. Undoing the normalization therefore gives exactly the asserted classification up to scale. $\square$

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